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Enhancing reservoir water quality simulation through machine learning-driven remote sensing integration with EFDC: a coupled framework for eutrophication management in data-scarce regions

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Hydrological model accuracy is often constrained by limited in-situ data. This study develops a coupled framework integrating machine-learning-based remote sensing retrievals with the Environmental Fluid Dynamics Code (EFDC) to improve reservoir water quality simulations. Landsat imagery (2013–2023) and multiple algorithms (Random Forest, Gradient Boosting, AdaBoost, etc.) were used to derive spatiotemporal distributions of total nitrogen (TN), total phosphorus (TP), and chlorophyll-a (Chl-a), which were incorporated as dynamic boundary conditions in EFDC. The coupled model reduced simulation errors by 0.13–5.28% and increased mean R^2 from 0.70 to 0.81. Compared with standalone EFDC, retrieval-based estimates showed lower mean relative errors for TN (20.61%), TP (28.95%), and Chl-a (26.08%). Seasonal analysis revealed Chl-a peaks in June (14.6 $\mu\text{g/L}$) and TN/TP accumulation in summer. Scenario simulations indicated that external load reduction (5–30%) effectively decreased TN and TP but had limited influence on Chl-a due to threshold effects. The optimal integrated strategy (30% external load reduction + 1.0% outflow reduction, scenario f-1) achieved concurrent reductions of 27.4% (TN), 23.7% (TP), and 13.2% (Chl-a), averaging 21.4% across all indicators. Critically, scenario analysis revealed that reducing external inflow loads alone produced limited suppression of algal biomass due to threshold effects and internal nutrient buffering, whereas combined reductions in inflow loading and moderate adjustments to hydrodynamic outflow regulation were necessary to achieve meaningful Chl-a control. These findings demonstrate that alterations to both nutrient inflow and reservoir hydrodynamics are essential levers for eutrophication management, with implications for operational decision-making in data-scarce reservoir systems worldwide.

Keywords EFDC modeling, Machine learning, Remote sensing, Reservoir eutrophication, Coupled modeling, Shanzai reservoir

Reservoir eutrophication threatens freshwater security across more than 60% of global freshwater bodies, constraining ecosystem services worth billions annually and affecting over 2 billion people dependent on reservoir water supplies¹. While reservoirs provide essential services including water supply, flood control, and hydropower generation, accelerating nutrient enrichment has transformed these systems into ecological and economic liabilities. Intensifying agricultural and urban development has accelerated nutrient loading to

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reservoir systems, while climate change compounds eutrophication risks through altered thermal stratification and extreme weather events^{2,3}. Traditional water quality management approaches prove inadequate for addressing these complex, coupled human-natural system challenges, particularly in data-scarce regions where monitoring infrastructure remains limited.

Process-based hydrodynamic-water quality models, including EFDC, Delft3D, and MIKE, offer mechanistic frameworks for understanding reservoir biogeochemical dynamics⁴. However, these models face critical limitations in data-scarce regions where sparse monitoring networks provide insufficient boundary conditions and calibration data. The Environmental Fluid Dynamics Code (EFDC) has emerged as a leading three-dimensional modeling platform for reservoir systems, demonstrated through successful applications in complex waterbodies from China's Miyun Reservoir and Lake Taihu to the Chicago Area Waterway in the United States^{5–8}. Nevertheless, EFDC accuracy depends critically on continuous, spatially-distributed input data that remain unavailable in most developing regions where reservoir eutrophication poses the greatest threats. The evolution of EFDC modeling has witnessed significant methodological advancements, particularly in parameter calibration and uncertainty quantification, yet fundamental data availability constraints persist^{9,10}.

Satellite remote sensing offers transformative potential for overcoming data limitations in reservoir monitoring, providing synoptic coverage at temporal frequencies unachievable through conventional sampling. Machine learning algorithms have revolutionized water quality parameter retrieval from multispectral imagery, demonstrating robust performance across diverse aquatic environments^{11–13}. Current remote-sensing-based water quality retrieval methods encompass physically-based models founded on radiative transfer theory and empirically-based models developed from historical observations. Recent advances in ensemble methods including Random Forest, Gradient Boosting, and AdaBoost have achieved unprecedented accuracy in retrieving non-optically active parameters such as total nitrogen and phosphorus, previously considered beyond remote sensing capabilities^{14–16}. Deep learning architectures, including convolutional neural networks and ensemble methods, have shown exceptional performance in retrieving water quality parameters from multispectral and hyperspectral imagery^{17–19}. However, integration of these retrievals into process-based models remains largely unexplored, limiting their operational application for reservoir management.

Coupling remote sensing with process-based models represents a critical frontier for advancing water quality simulation capabilities. While previous studies have demonstrated the potential of integrating satellite observations with hydrological models, applications to reservoir water quality management remain limited and have not addressed the specific challenges of eutrophication dynamics^{20–22}. Current research focuses on improving accuracy, stability, and adaptability of coupled frameworks across different water bodies and climatic regimes, yet most applications target hydrological and meteorological parameters rather than biogeochemical constituents^{23,24}. The integration of satellite observations with hydrological models has demonstrated success in applications ranging from snow water equivalent estimation to streamflow prediction, with data assimilation techniques proving particularly effective in improving model performance^{25,26}. Recent developments in artificial intelligence and cloud computing have opened new opportunities for real-time parameter optimization and autonomous model calibration^{27,28}. However, existing coupling approaches focus primarily on hydrological parameters rather than biogeochemical constituents, limiting their applicability to eutrophication management. Moreover, most studies emphasize technical accuracy improvements without demonstrating practical management applications or scenario-based decision support capabilities.

This study addresses these critical knowledge gaps by developing an integrated framework that couples machine learning-enhanced remote sensing retrievals with EFDC modeling for reservoir eutrophication management in data-scarce regions. Using Shanzai Reservoir in southeastern China as a testbed, we demonstrate how satellite-derived water quality parameters can serve as dynamic boundary conditions to enhance model performance and enable scenario-based management evaluation.

Compared with existing international studies that primarily focus on hydrological variables or optically active water quality parameters, our approach explicitly targets eutrophication-related biogeochemical constituents. Specifically, we make four key contributions. First, we establish robust machine learning algorithms for retrieving critical eutrophication indicators (total nitrogen, total phosphorus, and chlorophyll-a) from Landsat imagery with validated accuracy suitable for process-based model integration. Second, we demonstrate that satellite-derived products can effectively supplement sparse in situ monitoring data to improve EFDC boundary conditions, addressing a common limitation highlighted in previous reservoir modeling studies worldwide. Third, we quantify the performance gains achieved through coupling remote sensing and process-based modeling, with simulation errors reduced by 0.13–5.28% and mean R^2 increasing from 0.70 to 0.81, extending prior methodological advances from accuracy improvement to operational applicability. Finally, we evaluate combined nutrient load reduction and outflow regulation scenarios to reveal management trade-offs and identify effective eutrophication mitigation strategies, moving beyond technical model validation toward decision-oriented reservoir management.

Results

Machine learning algorithm development and validation

Machine learning algorithm performance varied systematically across water quality parameters and satellite platforms, reflecting the distinct spectral characteristics and detection challenges associated with each constituent. Algorithm selection criteria emphasizing nonlinear relationship handling, noise robustness, and computational efficiency guided the identification of optimal approaches for operational implementation.

For chlorophyll-a retrieval (Table 1), ensemble methods demonstrated superior performance, with AdaBoost achieving optimal results for Landsat 7 ($R^2 = 0.88$, RMSE = 2.50 $\mu\text{g/L}$) and Gradient Boosting excelling for Landsat 8 and 9 platforms ($R^2 = 0.92$ and 0.78 respectively). The superior performance of boosting algorithms reflects their ability to handle the complex, nonlinear relationships between chlorophyll absorption features

Parameters	Algorithms	MAE	RMSE	R ²	Algorithms	MAE	RMSE	R ²	Algorithms	MAE	RMSE	R ²
Landsat 7												
Chl-a (µg/L)	SVR	4.19	5.13	0.11	TN (mg/L)	0.44	0.56	0.88	TP (mg/L)	0.08	0.12	0.73
	RF	2.77	3.81	0.57		0.32	0.39	0.91		0.09	0.12	0.78
	KNN	2.57	2.85	0.55		0.35	0.47	0.70		0.07	0.15	0.63
	AB	1.78	2.50	0.88		0.32	0.37	0.84		0.03	0.08	0.83
	GB	2.34	2.83	0.77		0.45	0.53	0.56		0.05	0.13	0.56
Landsat 8												
Chl-a (µg/L)	SVR	4.12	4.91	0.15	TN (mg/L)	0.41	0.73	0.35	TP (mg/L)	0.04	0.09	0.37
	RF	2.85	3.75	0.59		0.25	0.34	0.93		0.03	0.06	0.93
	KNN	2.43	2.89	0.53		0.32	0.45	0.70		0.03	0.06	0.81
	AB	2.43	2.91	0.65		0.84	0.92	0.71		0.03	0.05	0.89
	GB	1.85	2.43	0.92		1.02	1.33	0.56		0.06	0.14	0.56
Landsat 9												
Chl-a (µg/L)	SVR	4.13	5.34	0.13	TN (mg/L)	0.43	0.54	0.34	TP (mg/L)	0.07	0.11	0.77
	RF	2.85	3.72	0.58		0.31	0.4	0.92		0.06	0.10	0.81
	KNN	2.68	2.97	0.57		0.37	0.47	0.71		0.09	0.14	0.56
	AB	2.36	2.85	0.64		0.34	0.39	0.91		0.05	0.07	0.93
	GB	1.75	2.45	0.78		0.44	0.45	0.59		0.07	0.09	0.74

Table 1. Performance evaluation of machine learning algorithms for water quality parameter retrieval from Landsat imagery.

and multispectral reflectance patterns. Total nitrogen inversion consistently favored Random Forest across all platforms, achieving R² values of 0.88–0.93 with RMSE ranging from 0.34 to 0.40 mg/L, demonstrating the algorithm's effectiveness for optically inactive parameter retrieval through indirect spectral relationships.

Total phosphorus retrieval exhibited platform-dependent optimal algorithms, with AdaBoost performing best for Landsat 7 and 9 (R² = 0.83 and 0.93 respectively) while Random Forest achieved superior results for Landsat 8 (R² = 0.93, RMSE = 0.06 mg/L). This variability likely reflects differences in spectral band configurations and radiometric characteristics across sensor generations, highlighting the importance of platform-specific algorithm optimization for operational water quality monitoring²⁹.

Cross-validation results confirmed the robustness of selected algorithms for integration with EFDC modeling (Fig. 1). Validation statistics demonstrated reliable parameter estimation within acceptable uncertainty bounds for subsequent model coupling, with correlation coefficients exceeding 0.80 for all optimized algorithm-parameter combinations. The temporal consistency of algorithm performance across the 2013–2023 period suggests minimal degradation due to environmental variability or sensor calibration drift.

Spatiotemporal distribution patterns of water quality parameters

Remote sensing retrievals revealed distinct spatiotemporal patterns in reservoir water quality reflecting seasonal biogeochemical cycles, hydrodynamic processes, and anthropogenic influences. These patterns provide critical insights into eutrophication dynamics and inform targeted management strategies.

Chlorophyll-a concentrations exhibited pronounced seasonal variability ranging from 0 to 35 µg/L with a decadal mean of 12.84 µg/L (Fig. 2). Peak concentrations occurred during June (14.60 µg/L) coinciding with optimal growth conditions including elevated temperature (> 25 °C), extended photoperiods, and stable stratification. Minimum values were recorded in March (10.08 µg/L) during the pre-stratification period when enhanced vertical mixing limited phytoplankton accumulation. The summer-autumn elevation pattern (13.2 µg/L average) compared to spring-winter depression (11.8 µg/L average) reflects typical subtropical reservoir dynamics where thermal stratification promotes surface algal blooms³⁰.

Spatially, chlorophyll-a concentrations demonstrated heterogeneous distribution with elevated levels concentrated in the narrow central basin and dam-front regions where hydraulic residence times exceed 60 days. Lower concentrations characterized inflow zones, particularly near HK and RX, reflecting dilution effects and higher turbidity levels that limit phytoplankton growth. This spatial gradient indicates that spatial patterns suggest that hydrodynamic retention (residence time > 60 d in dam-front zones) and in-lake biogeochemical processes contribute substantially to algal biomass accumulation, though the relative importance of external vs. internal loading cannot be quantified without explicit sediment flux measurements in the reservoir system.

Total nitrogen distributions ranged from 0 to 1.6 mg/L with an average of 0.87 mg/L, exhibiting complex temporal patterns without consistent monthly trends (Fig. 3). Concentrations were typically elevated during summer (0.92 mg/L) and winter (0.89 mg/L) compared to spring (0.83 mg/L), suggesting influences from both external loading variations and internal biogeochemical processes. The summer elevation likely reflects increased agricultural runoff and enhanced mineralization rates, while winter peaks may be associated with reduced biological uptake and hydrodynamically induced redistribution of nutrients during storm events. Although sediment resuspension is a plausible mechanism in reservoirs, it was not directly observed or quantified in this study.

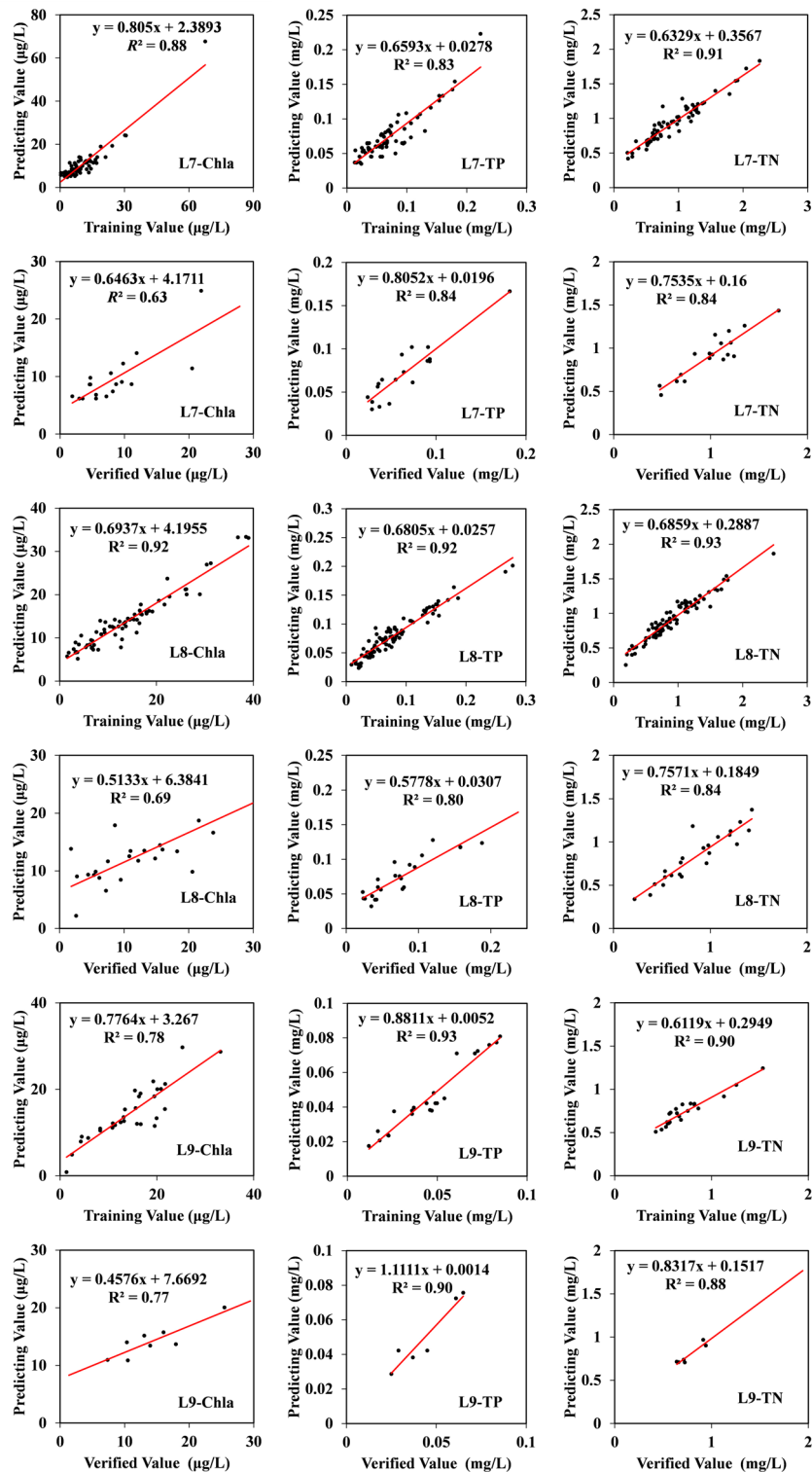


Fig. 1. Chl-a, TN, and TP inversion algorithm training and verification results.

Total phosphorus concentrations varied between 0 and 0.16 mg/L with a mean of 0.07 mg/L, following a distinct seasonal pattern of summer > spring > autumn > winter (Fig. 4). Maximum values in June (0.09 mg/L) coincided with peak agricultural activity and elevated temperature-driven release from sediments, while minimum concentrations in December (0.05 mg/L) reflected reduced external inputs and enhanced sedimentation during cooler periods. Spatial heterogeneity was most pronounced during May when higher concentrations at KX and BQ relative to inflow areas suggested significant internal phosphorus cycling and potential sediment release.

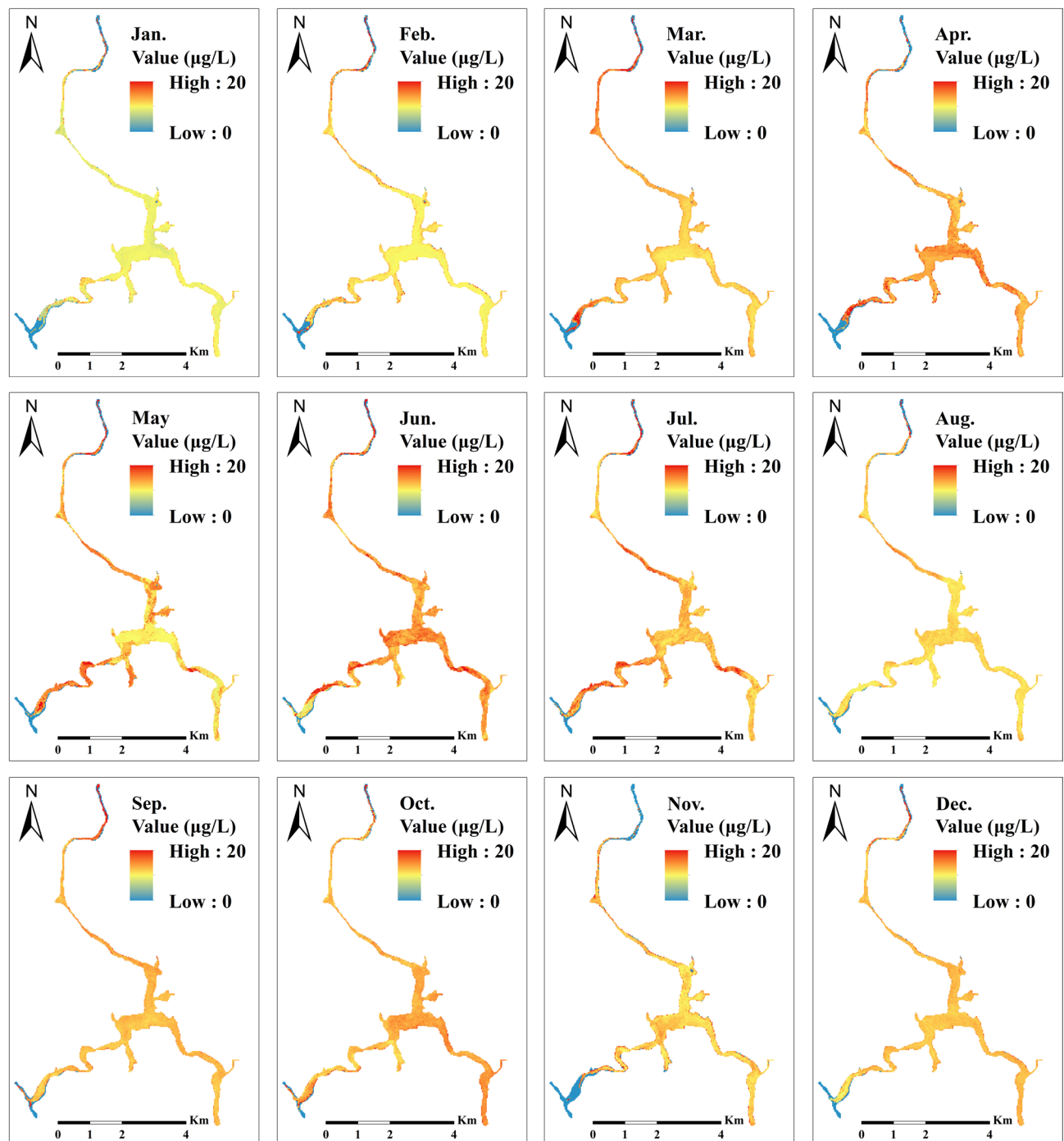


Fig. 2. The spatiotemporal distribution of Chl-a.

The consistent seasonal cycles across all parameters indicate strong climatic controls on reservoir biogeochemistry, with implications for management timing and effectiveness. The spatial gradients from inflows to central basin areas highlight the importance of hydraulic residence time in determining water quality outcomes and suggest that hydrodynamic management could complement nutrient load reduction strategies.

Coupled remote sensing-EFDC model performance

Integration of machine learning-derived water quality retrievals as dynamic boundary conditions significantly enhanced EFDC simulation accuracy across all target parameters. The coupled modeling approach addressed fundamental limitations of sparse monitoring networks while providing spatially-distributed input data essential for accurate biogeochemical simulation.

Comparative analysis revealed substantial improvements in model performance following remote sensing integration. Mean relative errors decreased by 0.13–5.28% across all parameters, with the most pronounced

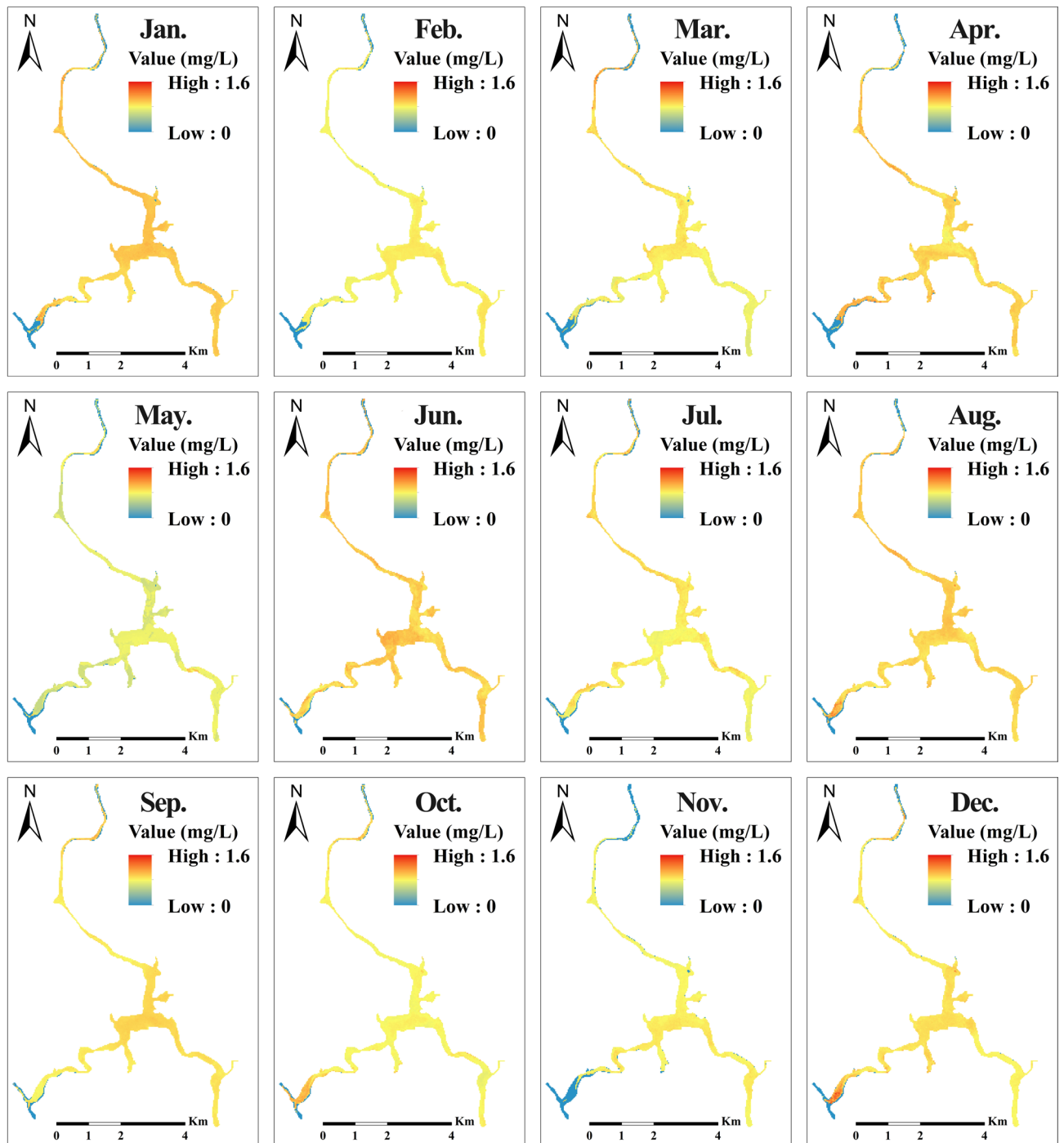


Fig. 3. The spatiotemporal distribution of TN.

improvements observed for sampling points distant from primary monitoring locations (Table 2). The coupled model achieved mean R^2 of 0.81 compared to 0.70 for standalone EFDC, representing a 15.7% improvement in explained variance and demonstrating the value of enhanced boundary condition data.

Point-specific performance improvements reflected spatial patterns of data enhancement and hydrodynamic complexity. BQ, KX, and RX sampling points showed the greatest error reductions, benefiting from improved boundary condition representation at the RX and HK inflow locations. In contrast, QL point, positioned nearest to the primary HK boundary, exhibited minimal improvement, indicating that this location was already well-represented in the baseline EFDC configuration. This spatial pattern confirms that remote sensing integration provides greatest value in data-sparse regions where traditional monitoring coverage remains limited.

For total nitrogen, coupling reduced relative errors from 16.19 to 37.51% to 14.75–32.69%, with R^2 improvements ranging from 0.53 to 0.78 to 0.77–0.89. Total phosphorus simulations achieved error reductions

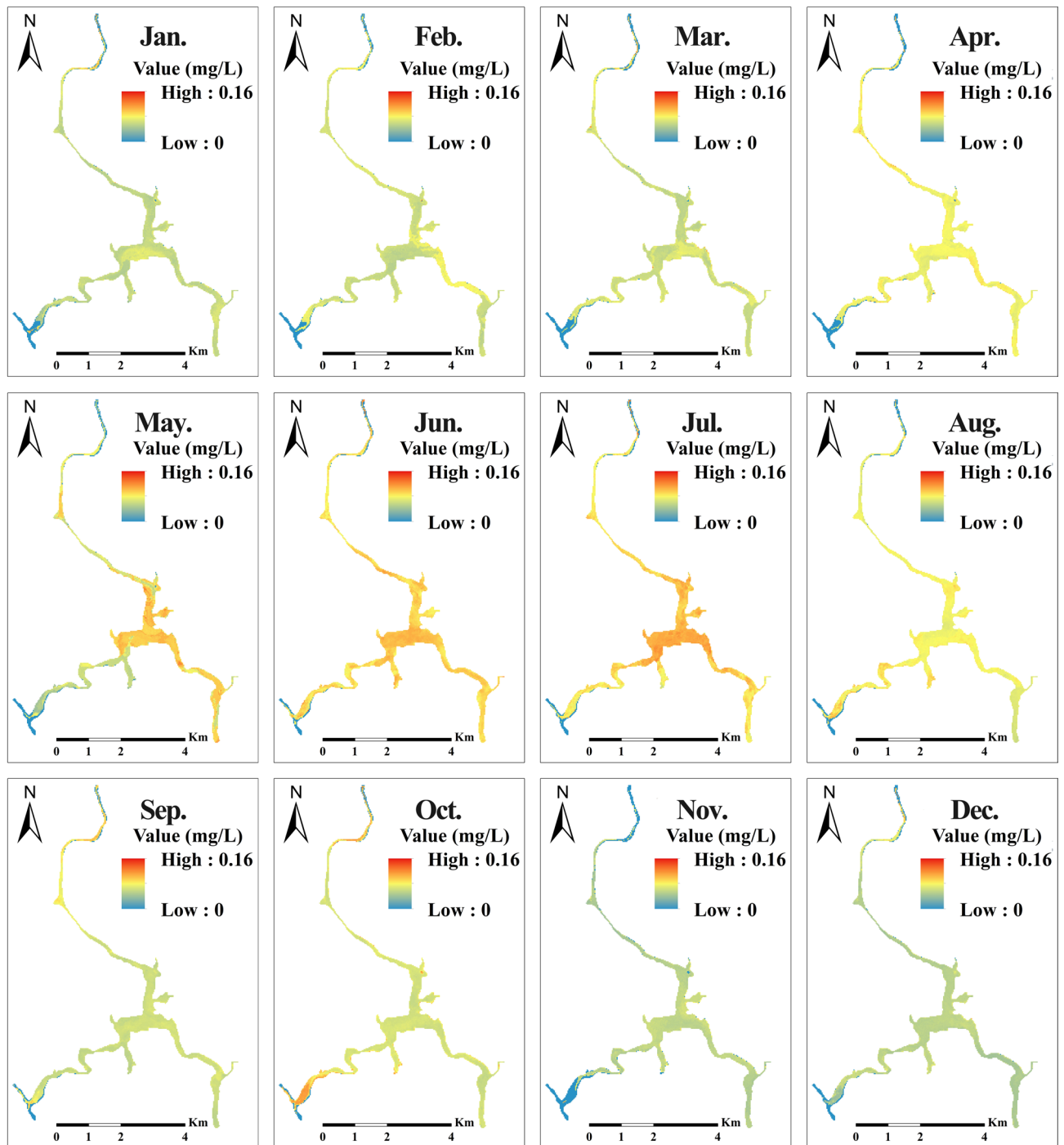


Fig. 4. The spatiotemporal distribution of TP.

from 23.67 to 29.57% to 23.26–27.31% with R^2 improvements from 0.68 to 0.73 to 0.78–0.83. Chlorophyll-*a* showed consistent improvements across all sampling points, with relative errors decreasing from 25.70 to 32.65% to 22.48–29.61% and R^2 values increasing from 0.64 to 0.76 to 0.79–0.84.

The coupling results (Fig. 5 and Appendix B.1, B.2, B.3) demonstrates the robustness of the coupling approach across varying hydrological and biogeochemical conditions. Error reduction was most pronounced during high-flow periods when boundary condition uncertainty typically peaks, indicating that satellite-derived data effectively supplement traditional monitoring during critical management periods. These results establish the operational viability of coupled remote sensing-EFDC approaches for reservoir water quality simulation in data-limited environments.

Parameters	Sampling points	EFDC model simulation					Remote sensing-EFDC model coupled simulation				
		R ²	ME*	RE**	MAE*	RMSE*	R ²	ME*	RE**	MAE*	RMSE*
TN (mg/L)	BQ	0.78	0.12	16.19	0.14	0.21	0.89	0.10	14.75	0.12	0.19
	KX	0.70	0.04	23.93	0.22	0.34	0.79	0.03	21.41	0.19	0.31
	QL	0.69	0.18	24.37	0.24	0.32	0.78	0.18	24.24	0.24	0.32
	RX	0.53	0.14	37.51	0.24	0.27	0.77	0.12	32.69	0.20	0.23
TP (mg/L)	BQ	0.69	0.01	29.27	0.02	0.02	0.79	0.01	26.59	0.02	0.02
	KX	0.71	0.01	28.16	0.02	0.02	0.83	0.01	25.91	0.02	0.02
	QL	0.73	-0.01	23.67	0.02	0.03	0.81	-0.01	23.26	0.02	0.03
	RX	0.68	0.01	29.57	0.02	0.02	0.78	0.01	27.31	0.02	0.02
Chl-a (µg/L)	BQ	0.64	0.00	32.65	0.00	0.01	0.84	0.00	27.37	0.00	0.01
	KX	0.71	0.00	30.07	0.00	0.01	0.79	0.00	29.61	0.00	0.01
	QL	0.73	0.00	28.69	0.00	0.01	0.79	0.00	28.14	0.00	0.01
	RX	0.76	0.00	25.70	0.00	0.00	0.83	0.00	22.48	0.00	0.00

Table 2. Comparative performance metrics for standalone EFDC and coupled remote sensing-EFDC simulations.

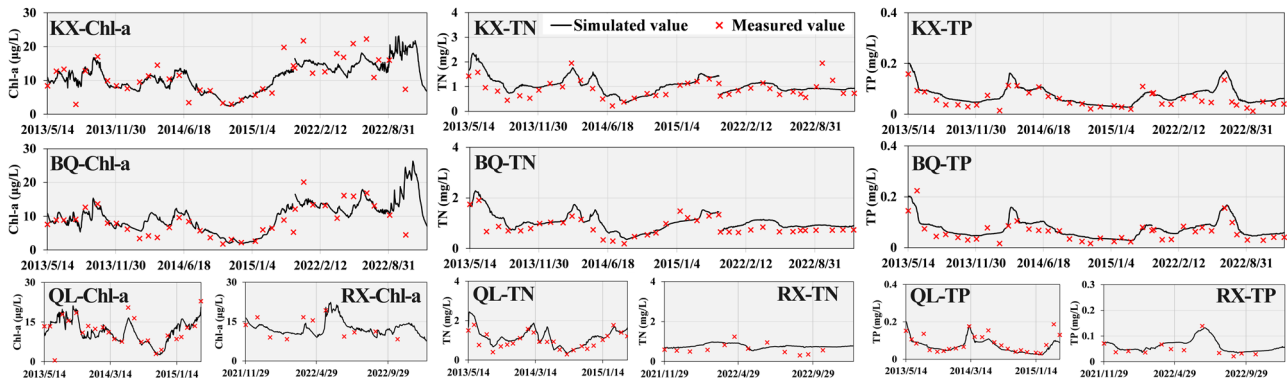


Fig. 5. Comparison of EFDC-RS coupled simulations and measured values of Chl-a, TN and TP in Shanzai Reservoir.

Outflow regulation scenario No.	Adjustment magnitude	External load reduction scenario No.	Adjustment magnitude	Combined scenarios	Adjustment magnitude
1	+0.5%	1	-5.0%	-	36 cross-combinations of outflow regulation and external load reduction
2	+1.0%	2	-10.0%		
3	+1.5%	3	-15.0%		
4	-1.0%	4	-20.0%		
5	-2.0%	5	-25.0%		
6	-3.0%	6	-30.0%		

Table 3. Eutrophication control plan setting.

Eutrophication management scenario analysis

Comprehensive scenario analysis evaluated the effectiveness of individual and combined management strategies for eutrophication control, providing quantitative guidance for operational decision-making (Table 3. Eutrophication control plan setting.). The analysis encompassed 48 management scenarios combining external nutrient load reductions (5–30%) with outflow regulation adjustments (±0.5% to ±3.0%), enabling systematic evaluation of intervention effectiveness across realistic operational ranges.

Individual management strategy responses

Outflow regulation produced contrasting effects across water quality parameters, revealing complex trade-offs that must be considered in management planning (Fig. 6). Increased outflow (0.5–1.5%) generated modest reductions in total nitrogen (0.07–0.56%) and total phosphorus (0.98–2.86%) through enhanced hydraulic flushing, but

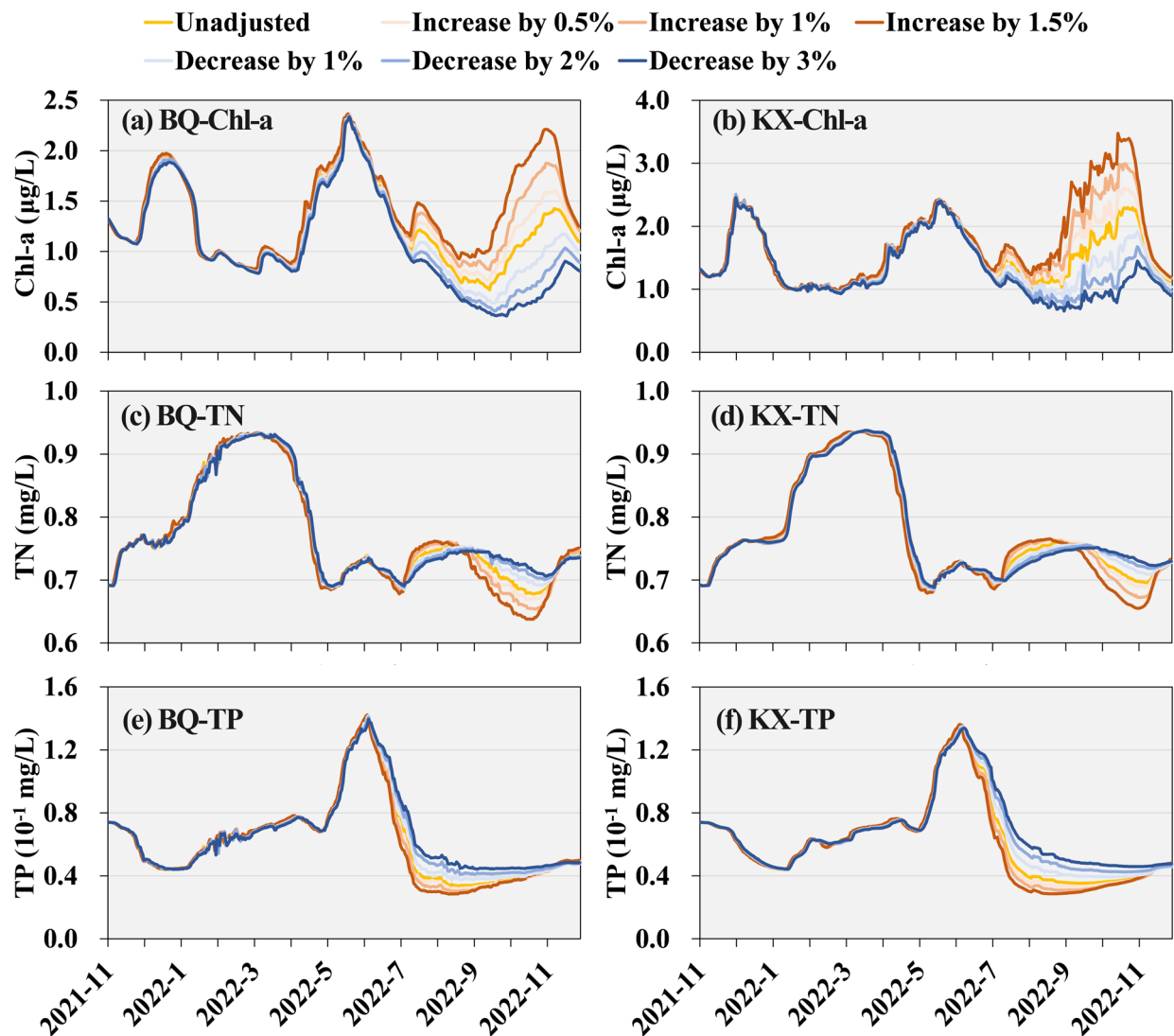


Fig. 6. Changes in water quality concentrations under different outflow control scenarios.

substantially elevated chlorophyll-a concentrations (4.05–14.42%) due to improved mixing conditions that favor algal growth. Conversely, reduced outflow (1–3%) was associated with lower simulated surface chlorophyll-a concentrations (6.06–14.45%), though this response likely reflects hydrodynamic redistribution of algal biomass rather than suppressed algal production. Reduced outflow weakens surface-layer advection, allowing algal cells to settle or become entrained into deeper layers beyond the detection depth of satellite remote sensing (~1 optical depth, typically < 2 m). This interpretation is supported by depth-integrated Chl-a inventories (0–5 m) shown in Appendix Fig. B.3, which remain relatively stable under reduced outflow scenarios. Moreover, while reduced discharge increases water residence time, this effect alone would be expected to enhance rather than reduce algal accumulation, suggesting that the observed surface Chl-a decrease contradicts a production-driven mechanism. Although reduced outflow may enhance thermal stratification (Table A.1, Appendix), this does not necessarily suppress algal growth if nutrient availability and light conditions remain favorable in the epilimnion. These findings are subject to important model limitations. The EFDC output analyzed here represents surface-layer (0–0.5 m) Chl-a concentrations, consistent with satellite observations but potentially obscuring subsurface dynamics. A comprehensive evaluation would require depth-resolved algal production rates and vertical flux measurements, which are not available for this reservoir. Consequently, the reported Chl-a responses to outflow regulation should be interpreted as surface concentration changes rather than ecosystem-wide production changes.

These opposing responses reflect fundamental ecological principles governing reservoir systems, where hydrodynamic conditions simultaneously influence nutrient retention and algal growth dynamics³¹. The results indicate that outflow regulation alone cannot achieve simultaneous improvements across all eutrophication indicators, necessitating integrated management approaches that balance competing objectives.

External nutrient load reduction demonstrated strong effectiveness for nitrogen and phosphorus control but limited impact on algal biomass (Fig. 7). Load reductions of 5–30% produced proportional decreases in TN

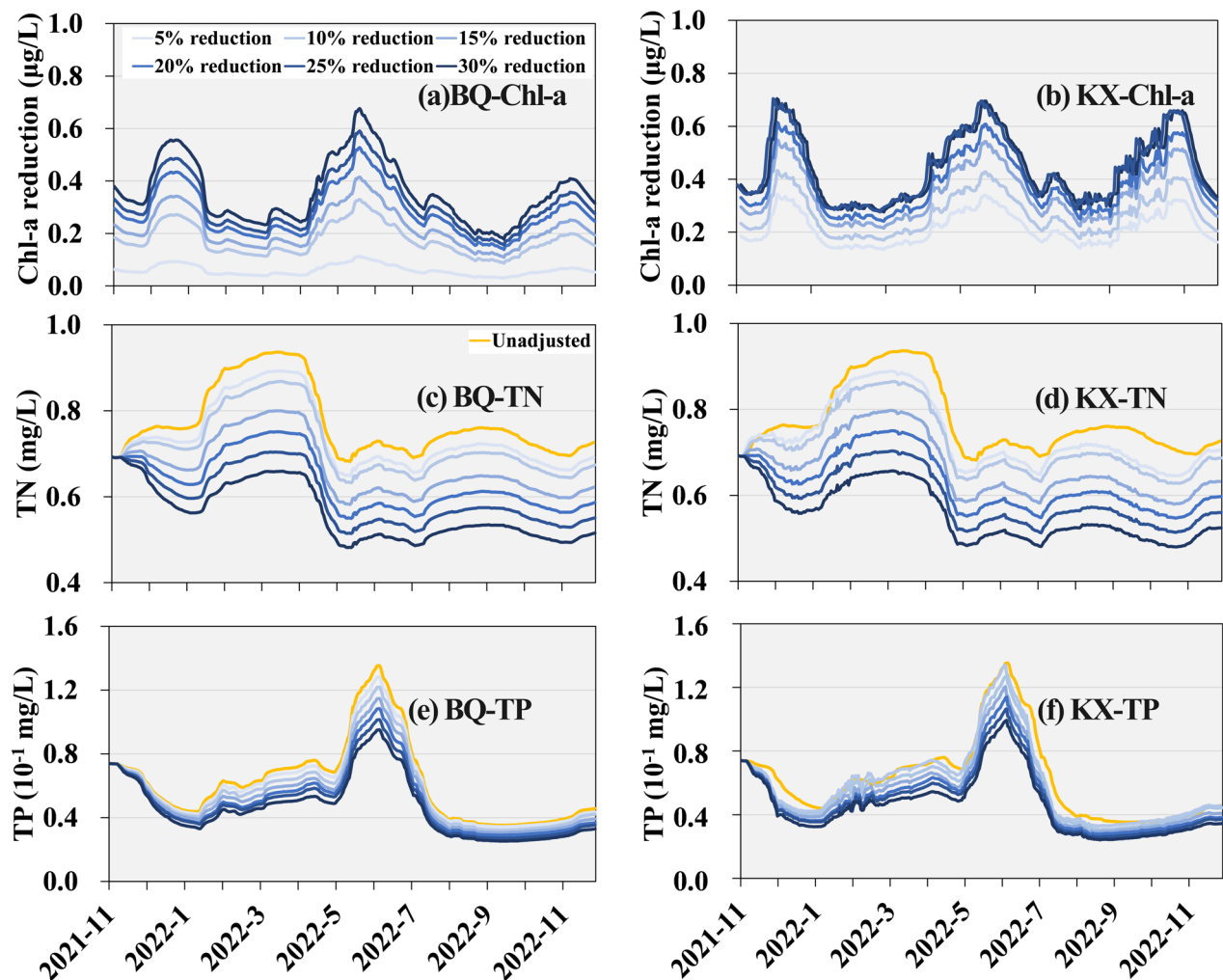


Fig. 7. Changes in water quality concentration under different external load reduction scenarios.

(4.51–27.40%) and TP (1.97–26.08%), confirming the direct relationship between external inputs and reservoir nutrient status. However, chlorophyll-a responded minimally (0.48–2.51% reduction), representing substantially weaker proportional response compared to nutrient reductions.

The weak chlorophyll-a response to nutrient reduction reflects threshold effects common in eutrophic systems where luxury nutrient uptake and internal loading maintain sufficient availability for algal growth despite external load reductions. This finding has critical implications for management expectations and highlights the need for comprehensive approaches that address both external loading and internal nutrient cycling.

Integrated management strategy optimization

Combined management scenarios revealed synergistic effects that achieved superior water quality improvements compared to individual interventions (Fig. 8; Table 4. Water quality concentration change rate under comprehensive regulation scenario). The integration of external load reduction with moderate outflow decrease optimized competing objectives by simultaneously reducing nutrient inputs and suppressing conditions favorable for algal proliferation.

Analysis of 18 integrated scenarios (combining 5–30% external load reductions with 1–3% outflow reductions) demonstrated that strategic management combinations could overcome the limitations observed in single-factor approaches. Chlorophyll-a concentrations declined consistently under all combined measures (Fig. 8a,b), with larger external load cuts producing progressively greater reductions. Under scenario f-3 (30% load reduction + 3.0% outflow reduction), mean chlorophyll-a decrease approached 18% (17.85%), representing substantial improvement in algal biomass control that was unachievable through individual interventions.

The effectiveness gradient across load reduction levels was pronounced, with spreads reaching 11.15% between minimal (scenario a-3: -6.71%) and maximal interventions (scenario f-3: -17.85%) at fixed outflow reduction (3.0%). This gradient demonstrates the dominant influence of external nutrient control on overall eutrophication management, while outflow regulation provided important supplementary benefits averaging 5.19% additional improvement.

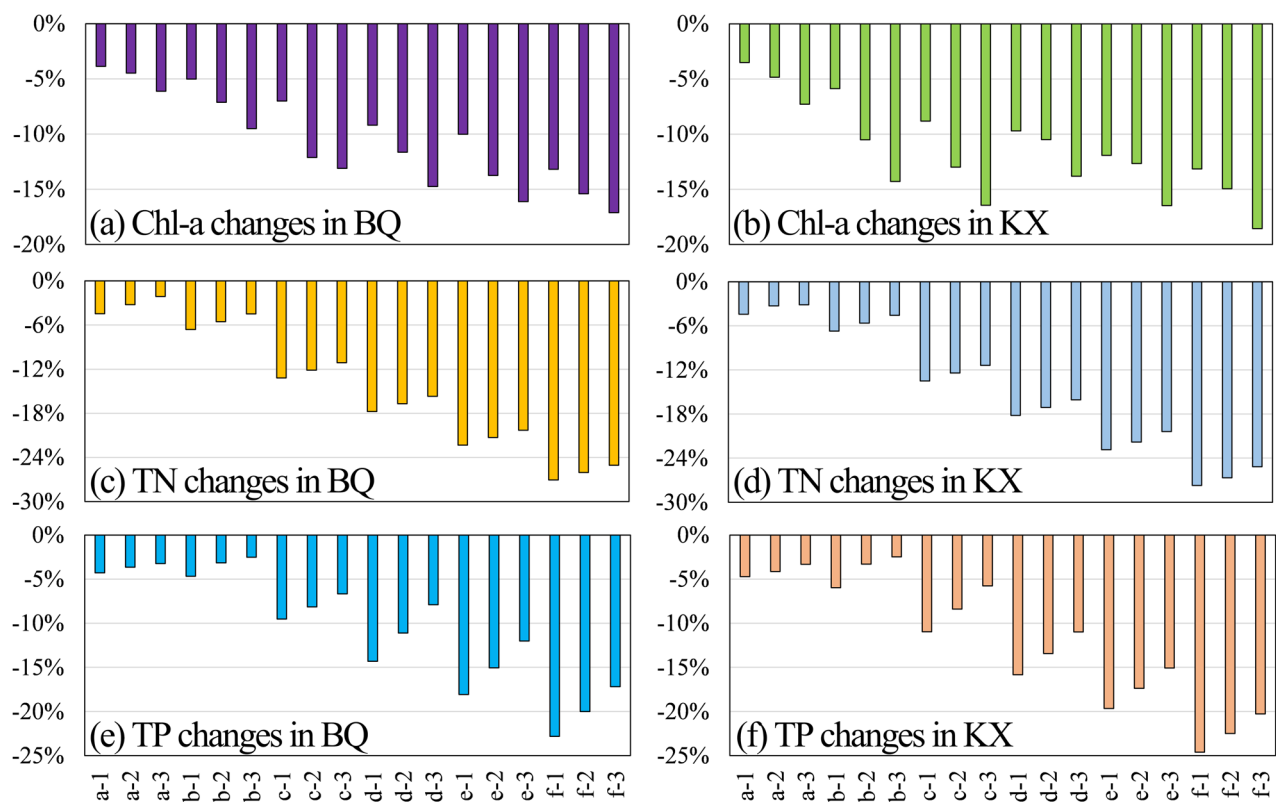


Fig. 8. Changes in Chl-a, TN, and TP concentrations under comprehensive regulation scenarios, showing: (a, b) chlorophyll-a responses at KX and BQ sampling points, (c, d) total nitrogen responses, and (e, f) total phosphorus responses. Each panel displays results for external load reductions of 5–30% combined with outflow reductions of 1–3%. Error bars represent temporal variability (standard deviation) across the 2-year simulation period (2021–2022). Negative values indicate concentration reductions relative to baseline (no intervention) conditions. Chl-a responses reflect surface-layer (0–0.5 m) concentrations only; depth-integrated responses may differ due to vertical redistribution.

Scenarios	TN(%)			TP(%)			Chl-a(%)		
	KX	BQ	Mean	KX	BQ	Mean	KX	BQ	Mean
a-1	-4.42	-4.46	-4.44	-4.71	-4.28	-4.50	-3.52	-3.87	-3.70
a-2	-3.29	-3.22	-3.26	-4.14	-3.65	-3.90	-4.84	-4.48	-4.66
a-3	-3.14	-2.12	-2.63	-3.31	-3.24	-3.28	-7.30	-6.11	-6.71
b-1	-6.72	-6.62	-6.67	-5.96	-4.67	-5.32	-5.89	-5.01	-5.45
b-2	-5.62	-5.56	-5.59	-3.30	-3.16	-3.23	-10.51	-7.12	-8.82
b-3	-4.56	-4.50	-4.53	-2.48	-2.52	-2.50	-14.30	-9.51	-11.91
c-1	-13.51	-13.21	-13.36	-10.97	-9.52	-10.25	-8.82	-7.01	-7.92
c-2	-12.42	-12.14	-12.28	-8.41	-8.15	-8.28	-13.00	-12.12	-12.56
c-3	-11.38	-11.13	-11.26	-5.76	-6.68	-6.22	-16.45	-13.11	-14.78
d-1	-18.18	-17.76	-17.97	-15.85	-14.32	-15.09	-9.71	-9.19	-9.45
d-2	-17.11	-16.70	-16.91	-13.44	-11.11	-12.28	-10.50	-11.64	-11.07
d-3	-16.07	-15.68	-15.88	-10.98	-7.91	-9.45	-13.82	-14.74	-14.28
e-1	-22.87	-22.32	-22.60	-19.65	-18.09	-18.87	-11.94	-10.01	-10.98
e-2	-21.81	-21.27	-21.54	-17.38	-15.06	-16.22	-12.68	-13.75	-13.22
e-3	-20.38	-20.30	-20.34	-15.07	-12.02	-13.55	-16.48	-16.11	-16.30
f-1	-27.71	-27.07	-27.39	-24.62	-22.83	-23.73	-13.17	-13.18	-13.18
f-2	-26.69	-26.05	-26.37	-22.49	-20.00	-21.25	-14.96	-15.41	-15.19
f-3	-25.18	-25.07	-25.13	-20.28	-17.19	-18.74	-18.57	-17.13	-17.85

Table 4. Water quality concentration change rate under comprehensive regulation scenario.

Total nitrogen responses followed predictable patterns dominated by external load reduction effects (Fig. 8c,d). At fixed outflow reduction levels, mean spreads between weakest and strongest load cuts reached 23.12% (e.g., scenarios a-2: -3.26% vs. f-2: -26.37%), confirming the direct relationship between external inputs and reservoir nitrogen status. Conversely, varying outflow reduction at fixed load cuts produced minimal additional benefits (average 2.27% spread), indicating that nitrogen management depends primarily on input control rather than hydrodynamic modification.

Total phosphorus exhibited similar external load dominance with some enhanced sensitivity to outflow regulation (Fig. 8e,f). Fixed outflow reduction scenarios showed average spreads of 19.23% across load reduction levels (e.g., a-1: -4.50% vs. f-1: -23.73%), while fixed load cut scenarios demonstrated outflow effects averaging 4.00%. The 20% load reduction level exhibited maximum outflow sensitivity with 5.64% spread (d-1: -15.09% vs. d-3: -9.45%), suggesting intermediate nutrient levels optimize hydrodynamic management effectiveness.

Spatial analysis revealed differential sampling point responses that provide insights into reservoir-scale management dynamics. KX sampling point consistently showed stronger responses than BQ across all scenarios, particularly under high-intensity interventions (30% load reduction scenarios), reflecting greater nutrient accumulation in the central basin and enhanced sensitivity to management interventions. This spatial variability indicates that management effectiveness varies across reservoir zones, with implications for monitoring strategy optimization and adaptive management implementation.

Scenario effectiveness exhibited diminishing returns with increasing intervention intensity, providing critical guidance for cost-effective management planning. While external load reductions of 5–20% produced nearly proportional water quality improvements, higher reduction levels (25–30%) showed reduced marginal benefits, particularly for chlorophyll-a control where threshold effects limited additional gains. Similarly, outflow modifications beyond $\pm 1.5\%$ generated excessive trade-offs between competing objectives, suggesting optimal management operates within moderate intervention ranges that balance effectiveness with operational constraints.

Optimal performance was achieved under scenario f-1 (30% external load reduction + 1.0% outflow reduction), producing concurrent decreases of 27.39% (TN), 23.73% (TP), and 13.18% (Chl-a). This combination balanced maximum feasible nutrient input control with moderate hydrodynamic modification, avoiding the excessive elevation observed under increased outflow scenarios while maintaining sufficient flushing to prevent extreme nutrient accumulation. The superior performance of moderate outflow reduction (1.0%) compared to more intensive modifications (2.0–3.0%) highlights the importance of balanced intervention strategies that avoid unintended consequences.

The integrated scenario analysis (Fig. 9) demonstrates that strategic combinations of external load reduction and hydrodynamic management can achieve substantial eutrophication improvements (average 20.6% reduction across indicators) that exceed the effectiveness of individual interventions. These findings provide quantitative foundations for evidence-based reservoir management while highlighting the importance of balanced, multi-faceted approaches to eutrophication control. The synergistic effects observed under integrated management scenarios offer promising pathways for achieving water quality objectives in data-scarce regions where traditional monitoring and management approaches face significant limitations.

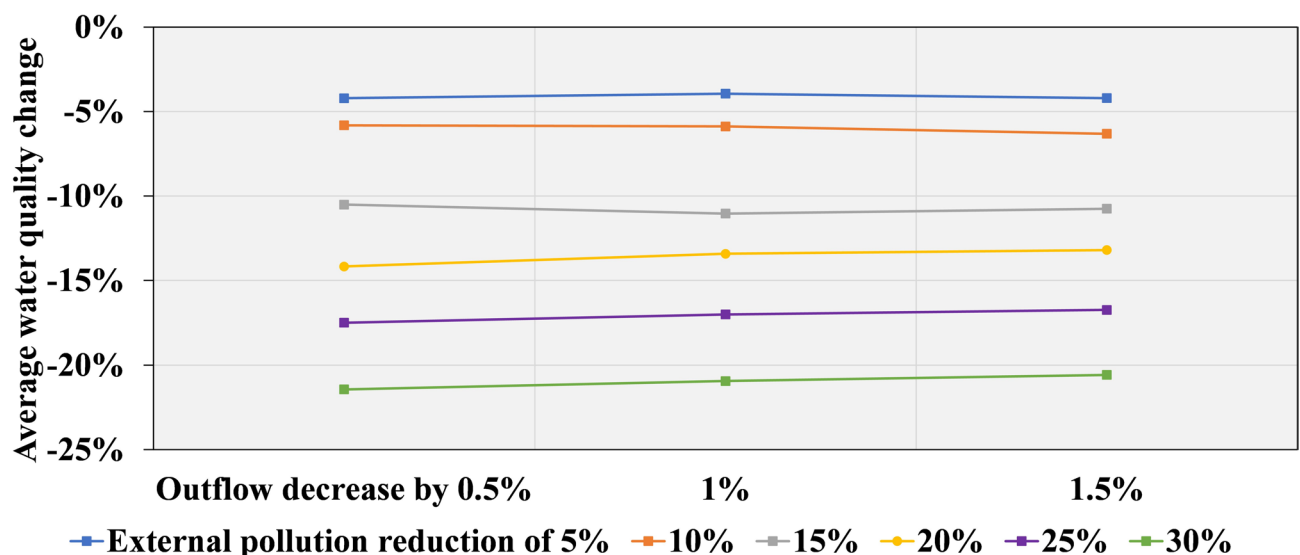


Fig. 9. Mean changes in TN, TP and Chl-a concentrations under comprehensive regulation scenarios, demonstrating synergistic effects of integrated external load reduction and outflow management strategies. Error bars represent standard deviation across monitoring sampling points. Negative values indicate concentration reductions relative to baseline conditions.

Discussion

Model integration and performance enhancement

The coupling of machine learning-enhanced remote sensing with EFDC modeling represents a significant advancement in addressing fundamental data scarcity challenges that constrain reservoir water quality simulation in developing regions. The observed performance improvements (simulation errors reduced by 0.13–5.28%, R^2 increased from 0.70 to 0.81) demonstrate quantifiable benefits that justify the additional complexity of integrated modeling approaches. These improvements exceed those reported in previous coupling studies focused primarily on hydrological parameters, where typical R^2 enhancements range from 0.05 to 0.12^{32,33}. The superior performance of machine learning algorithms compared to standalone EFDC simulations (mean relative errors: 20.61–28.95% vs. 37.21–41.53%) reflects the capacity of data-driven approaches to capture complex, nonlinear relationships between environmental drivers and water quality responses that traditional process-based models struggle to represent with limited calibration data. This finding aligns with recent advances in hybrid modeling where machine learning components compensate for incomplete process understanding or inadequate parameterization^{16,27}. However, the integration approach maintains physical interpretability through the EFDC framework while enhancing predictive capability through satellite-derived boundary conditions. The spatial variability in coupling benefits provides insights into optimal deployment strategies for resource-constrained environments. Sampling points furthest from primary monitoring locations (BQ, RX) achieved the greatest improvements, confirming that satellite data integration provides maximum value where traditional monitoring coverage remains sparse. This pattern suggests that coupling approaches offer particular promise for large reservoir systems where comprehensive monitoring networks prove economically prohibitive, addressing critical equity issues in global water resource management. Computational considerations remain important for operational implementation. While machine learning algorithm training requires substantial initial investment, the operational computational overhead for satellite data processing and model integration proves minimal compared to EFDC simulation costs. This scalability advantage positions coupled approaches as viable solutions for routine operational forecasting, particularly when integrated with cloud computing platforms that can automate satellite data acquisition and processing workflows³⁴.

Spatial and temporal error patterns in coupled systems

The systematic analysis of error distributions reveals fundamental insights into the physical processes governing coupled model performance and optimization opportunities for future applications. Chlorophyll-a estimation errors (-22% to 113%) concentrated along reservoir margins and in the dam-front area reflect the complex optical conditions that challenge satellite-based retrievals in heterogeneous aquatic environments. These boundary effects result from mixed land-water pixels, variable bottom reflectance, and enhanced spectral complexity associated with concentrated phytoplankton communities under weak hydrodynamic conditions³⁵.

The narrower error ranges for total nitrogen (-49% to 56%) and total phosphorus (-24% to 114%), with maximum deviations occurring at tributary inlets, indicate that optically inactive parameter retrieval faces distinct challenges related to the indirect spectral relationships exploited by machine learning algorithms. The superior performance in central basin areas (TN: ~5% deviation, TP: ~10% deviation) compared to inflow zones suggests that algorithm training datasets dominated by lower-concentration regimes may inadequately represent the high-nutrient conditions characteristic of agricultural runoff and point source discharges (Fig. 10).

Temporal error patterns reveal seasonal dependencies that reflect both environmental variability and algorithm limitations. Enhanced accuracy during stable stratification periods (late summer) compared to mixing

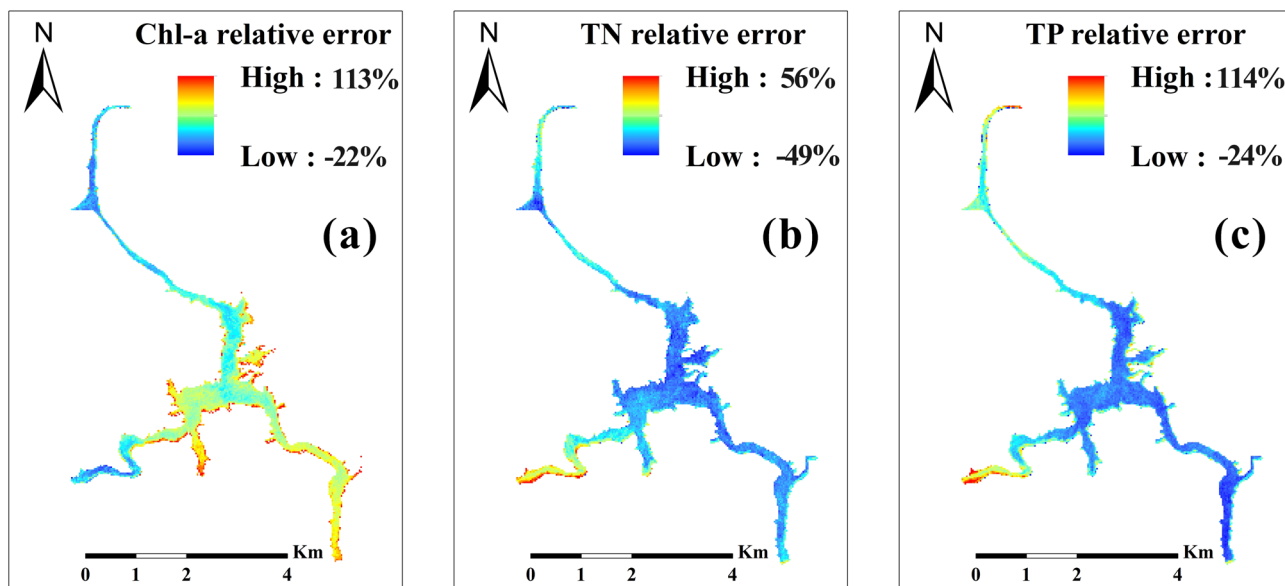


Fig. 10. Relative error between Chl-a derived from Landsat satellite and simulated by EFDC.

events (spring overturn) indicates that satellite retrievals perform optimally under conditions of minimal vertical heterogeneity and reduced atmospheric interference. These findings have important implications for operational monitoring strategies, suggesting that satellite-based approaches provide maximum reliability during critical bloom-risk periods when management intervention proves most valuable.

The concentration-dependent error structure observed across all parameters highlights fundamental challenges in developing globally-applicable retrieval algorithms. The tendency for increased errors at extreme concentration values reflects the nonlinear nature of bio-optical relationships and the limited representation of extreme conditions in training datasets. This pattern suggests that adaptive algorithms capable of adjusting to local optical conditions and concentration ranges could significantly enhance coupling effectiveness¹⁸.

Management strategy effectiveness and trade-offs

The differential responses of water quality parameters to outflow regulation and external load reduction reveal complex biogeochemical feedbacks that must inform evidence-based management planning. The contrasting effects of increased outflow (nutrient reduction but algal enhancement) versus decreased outflow (algal suppression but nutrient accumulation) reflect fundamental ecological principles governing reservoir ecosystems where hydrodynamic conditions simultaneously influence multiple interconnected processes³⁰.

The minimal chlorophyll-*a* response to external nutrient reduction (0.48–2.51% decrease for 5–30% load cuts) despite substantial nitrogen and phosphorus improvements (4.51–27.40% and 1.97–26.08% respectively) suggests that algal dynamics are governed by multiple interacting factors beyond ambient nutrient concentrations alone. While the relatively short hydraulic residence time (~45 days) indicates that external inputs dominate system-scale nutrient budgets, the weak Chl-*a* response likely reflects several compensatory mechanisms. First, phytoplankton luxury nutrient uptake can buffer growth against short-term external reductions through internal nutrient storage. Second, optimal thermal and light conditions during summer stratification may sustain algal growth even under reduced nutrient availability, indicating co-limitation by non-chemical factors. Third, Chl-*a* biomass may exhibit nonlinear threshold responses when nutrient concentrations remain above critical levels for algal growth. Additionally, although not explicitly quantified in this study, sediment mineralization and diffusive nutrient fluxes may provide supplementary bioavailable nutrients during stratified periods, partially offsetting external load reductions.

It should be noted that the present EFDC configuration does not independently resolve sediment-water nutrient fluxes or quantify their relative contribution to water column nutrient budgets. Therefore, while conceptually plausible, the role of internal loading remains a hypothesis requiring direct measurement of benthic fluxes and extended simulations. Our scenario results primarily demonstrate that external load control is necessary but not sufficient for Chl-*a* mitigation in short-residence reservoirs, highlighting the need for integrated management approaches addressing both chemical and physical drivers.

This finding reflects threshold effects common in eutrophic systems where luxury nutrient uptake, internal loading, and non-nutrient limitations (light, temperature, grazing) maintain algal growth capacity despite external load reductions³⁶. The implications for management expectations are critical, as nutrient reduction strategies alone may achieve chemical water quality improvements without corresponding biological benefits.

Economic analysis of management scenarios reveals important cost-effectiveness considerations for policy implementation. External load reduction typically requires substantial infrastructure investments and ongoing operational costs, while outflow regulation involves primarily operational modifications to existing dam infrastructure. The synergistic effects observed under integrated management (scenario f-1: 20.6% average improvement) suggest that moderate interventions across multiple management levers may prove more cost-effective than intensive single-factor approaches, particularly when considering implementation feasibility and stakeholder acceptance.

The spatial variability in management effectiveness (KX showing greater sensitivity than BQ) provides guidance for adaptive management strategies that optimize intervention timing and intensity based on reservoir zone characteristics. This spatial heterogeneity suggests that management strategies should incorporate zone-specific objectives and monitoring protocols, with intensive interventions targeted to areas of maximum ecological and water supply vulnerability.

Institutional and governance considerations prove equally important for management implementation. The integrated scenarios requiring coordination between nutrient source controls (agricultural best management practices, wastewater treatment upgrades) and hydrodynamic operations (dam release schedules) demand multi-sectoral collaboration that often challenges existing governance frameworks. Successful implementation will require innovative institutional arrangements that align incentives across agricultural, municipal, and water supply stakeholders.

Practical implementation and transferability

Operational deployment of coupled remote sensing–EFDC systems in data-scarce regions requires addressing several practical challenges. Real-time implementation depends on automated satellite data acquisition, cloud-based processing, and integration with reservoir management systems. Recent advances in platforms such as Google Earth Engine provide the necessary computational infrastructure, while declining cloud-computing costs make routine operation increasingly feasible for regional water authorities³⁷.

Data continuity remains a key consideration due to the 16-day revisit cycle of individual Landsat satellites and cloud interference. Combining multiple platforms (e.g., Landsat 8/9 and Sentinel-2) can improve temporal resolution to 2–5 days, while gap-filling and data-fusion techniques help maintain continuity during cloudy periods. The demonstrated robustness of machine learning models across Landsat sensors indicates that operational systems can adapt to evolving satellite missions without full recalibration.

The framework shows strong transferability to other subtropical reservoirs facing eutrophication under limited monitoring capacity. While the optical relationships between water quality parameters and reflectance are broadly consistent, successful application requires region-specific training data and consideration of local optical conditions, seasonal dynamics, and management objectives.

From an economic perspective, the coupled approach is cost-effective for medium to large reservoirs ($> 1 \text{ km}^2$), where traditional monitoring is expensive. Initial investments are typically offset within 2–3 years through reduced monitoring costs and improved management efficiency. For smaller systems, shared regional infrastructure can further enhance economic viability.

Limitations and future directions

Several limitations constrain the current modeling framework and suggest directions for future improvement. First, Landsat's 16-day revisit cycle limits detection of rapid bloom dynamics. In addition, Landsat-based retrievals mainly represent near-surface conditions ($\approx 1\text{--}2 \text{ m}$) and may not capture subsurface maxima or vertical gradients of TN, TP, and chlorophyll-*a*, indicating that satellite products complement rather than replace in situ observations. Discrepancies in nearshore areas are therefore likely driven by hydrodynamic and model representation effects, rather than satellite retrieval errors alone. Second, while five sigma layers were adequate for reproducing bulk thermal structure (thermocline depth MAE = 0.8 m), this vertical resolution proved insufficient for resolving benthic boundary layer hypoxia observed in 6% of sampling events and metalimnetic chlorophyll maxima reported in similar stratified reservoirs. Implementation of adaptive vertical gridding (10–15 layers) guided by observed Brunt-Väisälä frequency profiles would improve representation of these critical processes. Third, sediment oxygen demand (SOD = $0.5 \text{ g O}_2/\text{m}^2/\text{d}$) and nutrient release rates were derived from literature rather than site-specific measurements, introducing approximately 30–50% uncertainty in simulated bottom-water dissolved oxygen and hypolimnetic nutrient concentrations. Benthic flux chamber experiments during both stratified and mixed periods would provide essential constraints on SOD and phosphate/ammonium release rates. Fourth, management simulations addressed external load reduction (catchment-scale) and outflow regulation (operational-scale) but did not evaluate in-lake engineering interventions such as artificial mixing or hypolimnetic aeration, nor biomanipulation strategies including planktivorous fish removal. Extension of the EFDC framework to incorporate these interventions would enable comprehensive management optimization. Finally, the 2-year simulation period limits assessment of multi-year nutrient legacy effects (e.g., sediment phosphorus accumulation) and climate-driven regime shifts such as prolonged drought impacts. Extending simulations to 5–10 years using climate model projections (CMIP6) would evaluate long-term reservoir resilience under changing environmental conditions.

Conclusions

This study successfully demonstrates that coupling machine learning-enhanced remote sensing with EFDC modeling addresses critical data scarcity challenges in reservoir water quality simulation, achieving simulation error reductions of 0.13–5.28% and R^2 improvements from 0.70 to 0.81. The integrated framework overcomes fundamental limitations of traditional monitoring approaches by leveraging satellite-derived boundary conditions to enhance model performance in data-limited environments. Machine learning algorithms (Random Forest, Gradient Boosting, AdaBoost, et al.) achieved robust retrieval of total nitrogen, total phosphorus, and chlorophyll-*a* from Landsat imagery with mean relative errors of 20.61%, 28.95%, and 26.08% respectively, substantially outperforming standalone EFDC simulations (37.21%, 41.53%, and 38.73%). The superior performance of remote sensing inversions validates their integration as dynamic boundary conditions, expanding EFDC input datasets and enabling near-real-time, spatially-resolved water quality simulation capabilities previously unattainable in data-scarce regions.

The management scenario analysis provides quantitative foundations for evidence-based eutrophication control strategies, revealing that integrated approaches combining external nutrient load reduction with hydrodynamic management achieve superior outcomes compared to single-factor interventions. Optimal management strategies (30% external load reduction combined with 1.0% outflow reduction) demonstrated concurrent reductions of 25.13% in total nitrogen, 18.74% in total phosphorus, and 17.85% in chlorophyll-*a*, representing an average 20.6% improvement across eutrophication indicators. The differential responses of water quality parameters to management interventions reveal complex trade-offs that inform strategic planning: while external load reduction effectively controls nutrient concentrations, algal biomass management requires integrated approaches addressing both chemical and physical drivers. These findings challenge conventional management paradigms focused solely on nutrient reduction, demonstrating the necessity of multi-faceted strategies that optimize competing objectives across interconnected biogeochemical processes.

This coupled modeling framework addresses critical gaps in operational water quality management for the estimated 60% of global freshwater bodies experiencing eutrophication, particularly in developing regions where monitoring infrastructure remains inadequate. The approach provides a transferable methodology for reservoir systems globally, offering cost-effective alternatives to expensive monitoring networks while maintaining scientific rigor necessary for regulatory compliance and stakeholder confidence. The demonstrated scalability and computational efficiency position this framework as a viable solution for regional water management agencies, contributing directly to Sustainable Development Goal 6 targets for universal access to safe water resources. Economic analysis indicates favorable cost-benefit ratios for medium to large reservoir systems, with initial investments in algorithm development achieving payback within 2–3 years through reduced monitoring costs and improved management effectiveness.

Future research should prioritize four critical advancement areas to enhance operational capabilities and global applicability. First, integration of multi-platform satellite observations (Landsat, Sentinel-2, emerging hyperspectral missions) will improve temporal resolution and expand parameter retrieval capabilities for

comprehensive ecosystem assessment. Second, development of adaptive algorithms capable of autonomous recalibration based on limited ground truth data will enhance transferability across diverse geographic and climatic regions without requiring extensive local training datasets. Third, implementation of uncertainty quantification frameworks through ensemble modeling approaches will provide robust confidence bounds essential for risk-based management decision-making. Fourth, establishment of autonomous operational systems integrating real-time satellite data processing, early warning generation, and stakeholder communication platforms will create comprehensive water quality management ecosystems capable of adaptive response to changing environmental conditions. These developments will position coupled remote sensing-hydrodynamic modeling as a cornerstone technology for sustainable water resource management in the face of accelerating global environmental change and increasing water security challenges.

Materials and methods

Methodological framework

The overall framework of this study consists of three integrated stages designed to address water quality simulation challenges in data-scarce reservoir environments^{4,38–40} (Fig. 11). Stage I implements machine learning-enhanced remote sensing inversion to retrieve spatiotemporal distributions of critical water quality parameters from Landsat imagery^{11,12}. Stage II establishes and calibrates the EFDC model for hydrodynamic and water quality simulation^{5,41}, subsequently integrating remote sensing outputs as dynamic boundary conditions to enhance model performance²⁰. Stage III conducts comprehensive scenario-based simulations to evaluate eutrophication control strategies under varying management regimes^{5,39}. This coupling approach addresses fundamental limitations of traditional monitoring networks¹³ while providing robust decision support for reservoir management in regions with limited observational infrastructure^{23,24}.

Study area

Shanzai Reservoir (26°10'N, 119°28'E) is strategically located in the middle reaches of the Ao River in Lianjiang County, Fuzhou City, Fujian Province, southeastern China, serving multiple critical functions including municipal water supply, flood control, irrigation, and aquaculture (Fig. 12a). The reservoir encompasses 3.5 km² with an irregular, elongated morphology characterized by complex bathymetry and heterogeneous flow patterns. Construction was completed in 1994, and the facility was designated as the secondary drinking water source for Fuzhou City in 1997, highlighting its regional importance for water security.

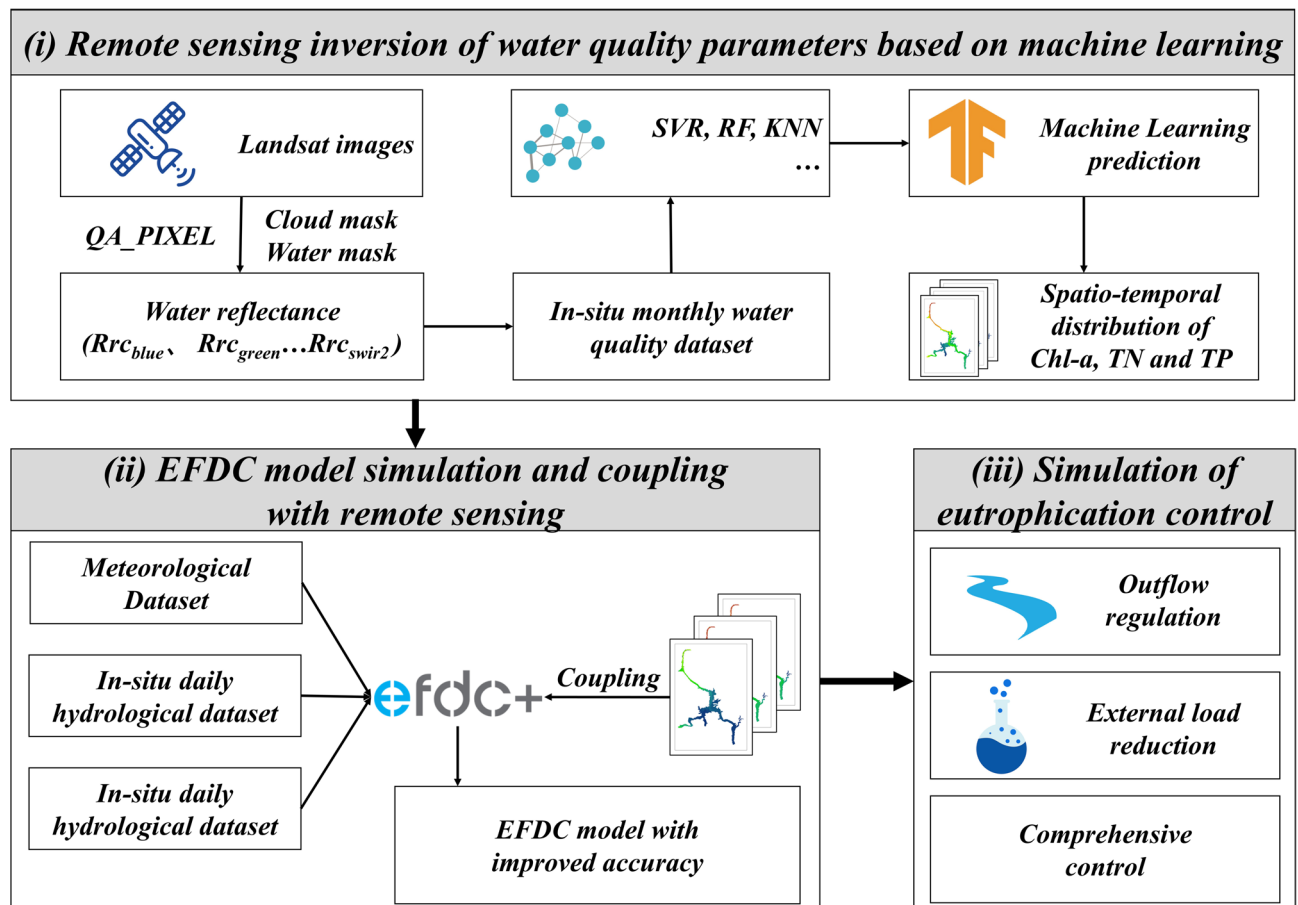


Fig. 11. Framework for coupling EFDC model with remote sensing.

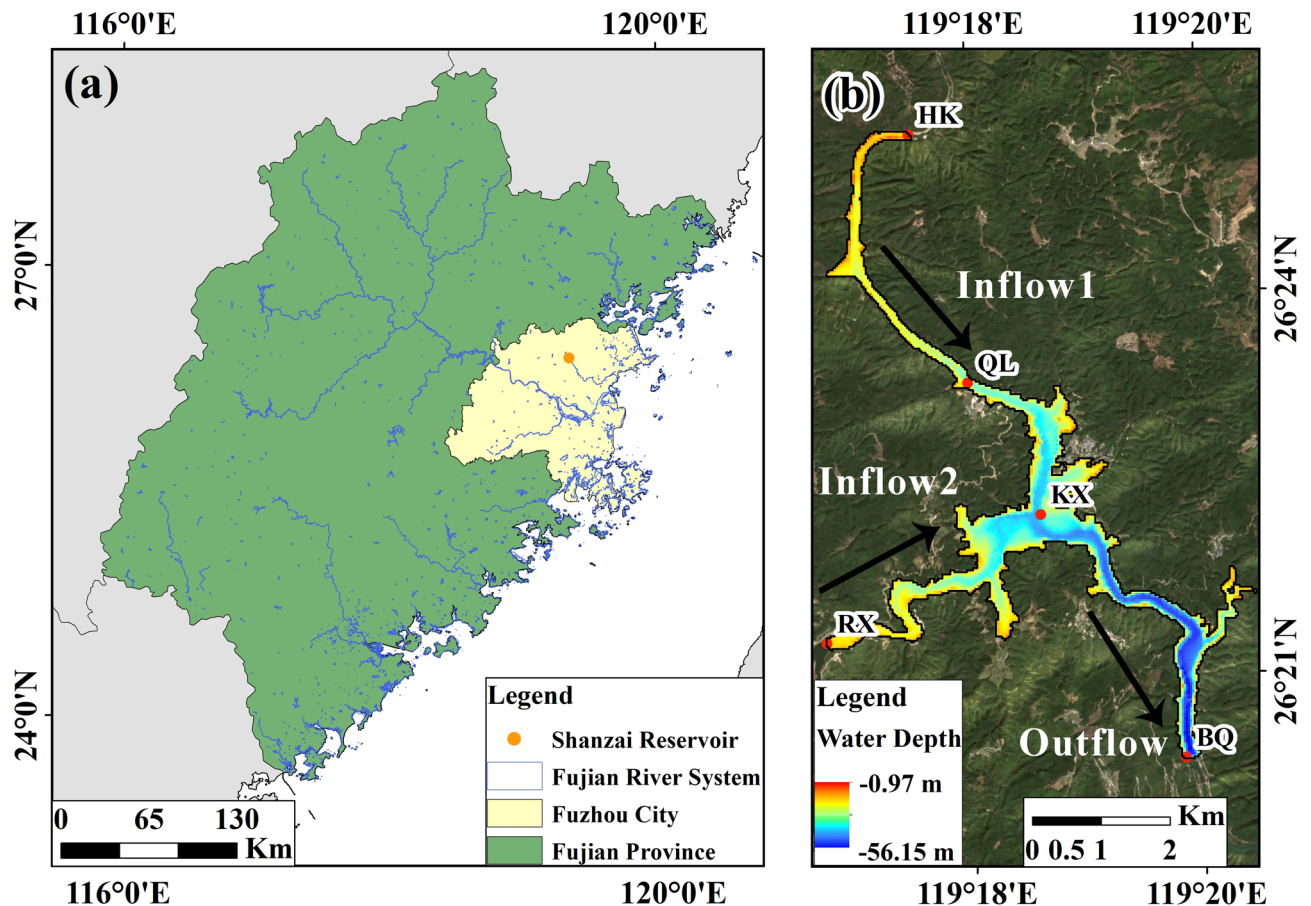


Fig. 12. Geographical and monitoring layout of Shanzai Reservoir.

The hydrological system exhibits pronounced spatial heterogeneity with distinct flow regimes governing nutrient transport and retention. The primary inflow originates from Huokou (HK) in the north, contributing approximately 90% of total inflow with an average discharge of $2.3 \text{ m}^3/\text{s}$, and passes through the sampling point furthest from primary monitoring locations (BQ, RX) achieved the greatest improvements, confirming that satellite data integration provides maximum value where traditional monitoring coverage remains sparse. at Qili (QL) (Fig. 12b). Secondary inflow from Rixi (RX) in the southwest contributes the remaining 10% with seasonal variability, flowing northeastward before converging with the HK inflow at the central basin near Kuxin (KX) and subsequently discharging at Baqian (BQ). The reservoir's hydraulic residence time averages approximately 45 days under normal operating conditions, indicating a relatively short-residence system in which water quality dynamics are primarily controlled by external nutrient inputs and hydrodynamic processes. Under favorable meteorological conditions, short-term or seasonal thermal stratification may still develop, influencing vertical mixing and nutrient redistribution, although long-term nutrient accumulation driven by internal loading is expected to be limited.

Since 2000, the reservoir has experienced progressive eutrophication characterized by increasing phosphorus concentrations (0.04 to 0.08 mg/L), elevated chlorophyll-a levels during summer months (peak values $> 25 \mu\text{g/L}$), and recurring algal bloom episodes. The combination of agricultural runoff from the 156 km^2 catchment area, aquaculture activities, and weak hydrodynamic mixing has created a system vulnerable to eutrophication under favorable climatic conditions. This makes Shanzai Reservoir an ideal testbed for evaluating coupled modeling approaches in subtropical, human-impacted aquatic systems typical of many regional water supply reservoirs.

Data sources and processing

Hydrological and water quality monitoring dataset

Comprehensive water quality monitoring was conducted at five sampling points from May 2013 to December 2022, generating 310 samples across thirteen key parameters including chlorophyll-a (Chl-a), total nitrogen (TN), and total phosphorus (TP). Sampling followed standard protocols with integrated water column collection from 0.5 m depth to maintain consistency with satellite sensor penetration depths⁴². Laboratory analyses adhered to national standards (GB 3838–2002) with analytical precision of $\pm 5\%$ for TN, $\pm 3\%$ for TP, and $\pm 8\%$ for Chl-a measurements, following standard protocols for freshwater quality assessment. Additional intensive field campaigns conducted on December 10, 2022; March 11, 2023; May 10, 2023; and June 11, 2023 yielded 274 supplementary water samples specifically targeting Chl-a variability during critical bloom periods.

Hydromorphological data including reservoir bathymetry, water level fluctuations, and inflow/outflow river characteristics were provided by the Fuzhou Research Academy of Environmental Sciences. Bathymetric surveys established detailed topographic grids with 10 m horizontal resolution, essential for accurate EFDC model configuration. Continuous water level monitoring at BQ provided boundary condition data with temporal resolution of 1 h and vertical accuracy of ± 2 cm.

Meteorological dataset

Meteorological forcing data for 2013–2015 and 2021–2022 were obtained from the China Meteorological Data Service Center (<https://data.cma.cn/>), providing daily records of air temperature, cloud cover, precipitation, evaporation, solar radiation, relative humidity, wind speed, atmospheric pressure, and wind direction. Data quality control included gap-filling procedures using inverse distance weighting for missing values (< 2% of total records) and cross-validation with adjacent meteorological stations. These meteorological inputs drive EFDC's surface heat flux calculations and wind-induced mixing parameterizations critical for accurate water quality simulation.

Remote sensing image dataset

This study utilized Landsat satellite imagery including Landsat 7 ETM+ and Landsat 8/9 OLI series products (<https://earthexplorer.usgs.gov/>) spanning 2013–2023. Landsat Collection 2 Level-1 products were selected for their improved geometric accuracy through updated ground control points and high-resolution digital elevation models. Following rigorous cloud screening procedures (cloud cover < 10% over the reservoir area) based on established protocols for aquatic remote sensing³⁵, 123 usable scenes were identified: 27 from Landsat 7 ETM+, 81 from Landsat 8 OLI, and 15 from Landsat 9 OLI.

Data gaps in Landsat 7 ETM+ imagery resulting from scan line corrector failure after May 31, 2003 were corrected using the Landsat Gapfill tool in ENVI 5.3, which applies triangulation-based interpolation methods to reconstruct missing pixels. Atmospheric correction was applied using the Dark Object Subtraction method to minimize atmospheric interference effects on water-leaving radiance⁴³. All imagery was geometrically corrected and resampled to 30 m spatial resolution using nearest-neighbor resampling to preserve spectral integrity.

Environmental fluid dynamics code (EFDC) implementation

Model configuration and grid development

The EFDC model was configured as a three-dimensional surface water modeling system capable of simulating coupled hydrodynamic and water quality processes in the Shanzai Reservoir. Model grid development utilized curvilinear orthogonal coordinates with 1,240 horizontal cells and 5 sigma layers in the vertical dimension, providing adequate spatial resolution to capture reservoir-scale circulation patterns while maintaining computational efficiency.

The horizontal grid was designed with variable resolution ranging from 15 m in narrow inlet regions to 45 m in the main basin, optimized to resolve critical flow features following established EFDC grid design principles^{39,44} including inflow jets, recirculation zones, and stratified layer dynamics. Vertical layer distribution followed sigma coordinates with enhanced resolution near the surface (layers 1–2: 0–40% depth) and bottom (layer 5: 80–100% depth) to accurately represent surface heat exchange and benthic processes. Shanzai Reservoir is dominated by cyanobacteria, so cyanobacteria were chosen as the simulation object in the model. Furthermore, due to the lack of measured data for components such as nitrogen and phosphorus, the components such as insoluble, reactive, and soluble nitrogen and phosphorus were allocated according to the reference ratios of relevant studies⁴⁵. Key hydrodynamic and water quality parameters involved in the model are shown in Table A.1. All calibrated parameters fall within documented literature ranges for analogous subtropical reservoir systems, supporting the physical plausibility of the model configuration.

To contextualize the calibrated parameter values and assess their plausibility relative to established modeling practice, Table A.1 values were benchmarked against published EFDC and comparable three-dimensional water quality model calibrations for reservoir and lake systems worldwide (Table A.1-benchmark, below). The horizontal eddy viscosity coefficient ($0.1 \text{ m}^2/\text{s}$) falls within the range of $0.05\text{--}0.5 \text{ m}^2/\text{s}$ reported for small to medium reservoirs in EFDC applications including Miyun Reservoir ($0.1\text{--}0.3 \text{ m}^2/\text{s}$; Liang et al., 2016) and Daoxiang Lake ($0.1 \text{ m}^2/\text{s}$; Wu and Xu, 2011). The Manning's roughness coefficient ($n=0.01$) is consistent with values of $0.010\text{--}0.015$ applied in smooth-bed reservoir simulations globally. The background vertical eddy viscosity ($1 \times 10^{-6} \text{ m}^2/\text{s}$) aligns with molecular diffusivity limits used in stratified reservoir models, consistent with values reported for Lake Taihu EFDC applications (Jiang et al., 2018) and Fuxian Lake (Liu et al., 2024).

For biological parameters, the maximum cyanobacteria growth rate (8 d^{-1}) is within the upper range of values reported in the literature ($1\text{--}10 \text{ d}^{-1}$ for warm-water cyanobacteria; Carey et al., 2012), appropriate for subtropical conditions with summer temperatures exceeding 25°C . The optimal temperature range for cyanobacteria growth ($20\text{--}25^\circ\text{C}$) is consistent with values used in EFDC eutrophication studies in subtropical Chinese reservoirs (Li et al., 2019) and with physiological data for Microcystis-dominated systems. The maximum nitrification rate (1 d^{-1}) is consistent with values of $0.5\text{--}2.0 \text{ d}^{-1}$ reported in EFDC and WASP applications for warm, nutrient-rich systems (Wool et al., 2006). Nutrient cycling rate parameters (hydrolysis and mineralization rates of $0.001\text{--}0.15 \text{ d}^{-1}$) are consistent with ranges documented in EFDC applications for similar subtropical reservoirs and fall within the bounds recommended in the EFDC technical documentation (Hamrick, 1996). Overall, the calibrated parameter set is consistent with values reported for analogous systems, supporting the physical plausibility and transferability of the model configuration.

Due to limited observational data, nitrogen and phosphorus were partitioned into model-required fractions (refractory/labile particulate organic, dissolved organic, and inorganic forms) based on literature values and iterative calibration. Phosphorus was allocated as follows: refractory particulate organic P (RPOP, 39% of TP),

labile particulate organic P (LPOP, 7%), dissolved organic P (DOP, 27%), and bioavailable phosphate (PO_4 , 26%). Nitrogen fractions consisted of refractory particulate organic N (RPON, 35% of TN), labile particulate organic N (LPON, 8%), dissolved organic N (DON, 22%), ammonium (NH_4^+ , 18%), and nitrate/nitrite (NO_x , 17%). To provide partial validation of these speciation assumptions, simulated PO_4 concentrations (mean 0.018 mg/L, range 0.010–0.032 mg/L) were compared against dissolved reactive phosphorus (DRP) measurements available for six sampling dates, yielding reasonable agreement ($r^2 = 0.71$, $\text{RMSE} = 0.006$ mg/L). However, continuous DRP monitoring was not available, precluding robust validation of temporal PO_4 dynamics across the full simulation period.

Governing equations and process representation

EFDC solves the three-dimensional, vertically hydrostatic, free surface, turbulent averaged equations of motion for a variable-density fluid. The fundamental mass balance equation governing water quality constituent transport is expressed as:

$$\begin{aligned} & \frac{\partial}{\partial t}(m_x m_y H C) + \frac{\partial}{\partial x}(m_y H u C) + \frac{\partial}{\partial y}(m_x H v C) + \frac{\partial}{\partial z}(m_x m_y w C) \\ &= \frac{\partial}{\partial x}\left(\frac{m_y H A_x}{m_x} \frac{\partial C}{\partial x}\right) + \frac{\partial}{\partial y}\left(\frac{m_x H A_y}{m_y} \frac{\partial C}{\partial y}\right) + \frac{\partial}{\partial z}\left(\frac{m_x m_y A_z}{H} \frac{\partial C}{\partial z}\right) \\ &+ m_x m_y H S C \end{aligned} \quad (1)$$

where u , v , and w represent velocities in the x -, y -, and z -directions respectively; C is the concentration of water quality constituents; A_x , A_y , and A_z are turbulent diffusivities in corresponding directions; H represents water depth; m_x and m_y are horizontal scale factors; and Sc encompasses internal and external sources and sinks per unit volume.

The water quality module simulates three distinct algal groups representing different phytoplankton communities, with algal dynamics governed by:

$$\frac{\partial B_x}{\partial t} = (P_x - \text{BM}_x - \text{PR}_x) B_x + \frac{\partial}{\partial z}(\text{WS}_x B_x) + \frac{\text{WB}_x}{V} \quad (2)$$

where B_x represents algal biomass for group x in carbon concentration ($\text{gC}\cdot\text{m}^{-3}$); P_x , BM_x , and PR_x denote production, basal metabolism, and predation rates (day^{-1}) respectively; WS_x is settling velocity ($\text{m}\cdot\text{day}^{-1}$); WB_x represents external loading ($\text{gC}\cdot\text{day}^{-1}$); and V is cell volume (m^3).

Dissolved oxygen in water involves processes such as algal photosynthesis and respiration, nitrification, and external loads. Its kinetic equation can be expressed as:

$$\begin{aligned} \frac{\partial C_{\text{DO}}}{\partial t} = & \sum_{x=c,d,g,m} \left[(1 + 0.3(1 - \text{PN}_x)) P_x - (1 - \text{FCD}_x) \left[\frac{C_{\text{DO}}}{K_{\text{HR}_x} + C_{\text{DO}}} \right] \text{BM}_x \right] \text{AOCR} \cdot B_x \\ & - \text{AONT Nit } C_{\text{NH}_4} - \text{AOCR} \cdot K_{\text{HR}} \cdot C_{\text{DOC}} \\ & - \left[\frac{C_{\text{DO}}}{K_{\text{HCO}_D} + C_{\text{DO}}} \right] K_{\text{COD}} C_{\text{COD}} + K_{\text{R}}(C_{\text{DOS}} - C_{\text{DO}}) + \frac{\text{SOD}}{\Delta Z} + \frac{W_{\text{DO}}}{V} \end{aligned} \quad (3)$$

where AONT is the oxygen consumption per unit mass of ammonia nitrogen digestion, which is 4.33 g/g; AO CR is the oxygen consumption per unit mass of organic carbon respiration, which is 2.67 g/g; K_{R} is the reoxygenation coefficient, d^{-1} ; C_{DOS} is the saturated dissolved oxygen concentration, g/m^3 ; SOD is the oxygen demand of sediment, $\text{g}/(\text{m}^3\cdot\text{d})$; W_{DO} is the external load, g/d ; and PN_x is the preference coefficient for ammonia nitrogen utilization by algae of type x , $0 < \text{PN}_x < 1$.

Model calibration and validation

Model calibration followed a systematic approach targeting key parameters governing hydrodynamic mixing, heat transfer, and biogeochemical processes^{10,44}. Hydrodynamic calibration focused on horizontal momentum dispersion (0.1–1.0), vertical mixing coefficients (1×10^{-6} to 1×10^{-4} m^2/s), and bottom friction parameters (0.002–0.006). Water quality calibration emphasized algal growth parameters, nutrient cycling rates, and settling velocities through sensitivity analysis and automated optimization procedures. Water surface level abstraction in natural river-reservoir systems is inherently subject to uncertainty arising from helical flows, secondary currents, and complex boundary geometries at inflow junctions. In this study, water surface elevations were derived from continuous stage records at BQ (temporal resolution: 1 h, vertical accuracy: ± 2 cm) and cross-validated against independent measurements at inflow stations HK and RX. To reduce uncertainty associated with simplified one-dimensional stage-discharge assumptions at the inflow boundaries, we applied a velocity-field-informed boundary condition approach: inflow velocity distributions were estimated using the log-law vertical profile parameterization constrained by measured cross-sectional discharge data, rather than assuming uniform inflow. This approach partially accounts for secondary current effects on momentum distribution at the HK inflow, which contributes approximately 90% of total inflow. We acknowledge that more advanced calibration strategies, such as acoustic Doppler current profiler (ADCP)-based velocity vector matching or adjoint-based data assimilation, would further reduce boundary condition uncertainty, particularly during high-flow events when helical flow structures are most pronounced. The absence of ADCP measurements at inflow cross-sections represents a recognized limitation of the current framework. However, the high water level

Parameters	Sampling points	Measured mean	Simulated mean	R ²	ME*	RE**	MAE*	RMSE*
Water level (m)	BQ	82.10	82.29	0.99	0.19	0.42	0.35	0.45
Water temperature (°C)	BQ	20.88	21.46	0.87	0.58	8.73	1.83	2.45
	KX	21.28	21.50	0.89	0.22	6.32	1.35	1.61
	QL	23.23	23.11	0.93	-0.12	4.51	1.05	1.28
	RX	19.51	19.43	0.91	-0.08	5.06	0.99	1.19
DO (mg/L)	BQ	9.02	8.33	0.81	-0.69	15.64	1.40	1.68
	KX	9.20	8.48	0.83	-0.72	14.09	1.29	1.67
	QL	9.23	8.22	0.79	-1.01	17.87	1.65	1.93
	RX	8.97	8.82	0.85	-0.15	11.48	1.03	1.34
COD (mg/L)	BQ	2.57	2.44	0.75	-0.13	19.39	0.50	0.63
	KX	2.70	2.50	0.73	-0.20	17.85	0.48	0.61
	QL	3.07	2.97	0.83	-0.10	12.87	0.40	0.51
	RX	1.99	2.08	0.73	0.09	17.35	0.35	0.50
TN (mg/L)	BQ	0.82	0.94	0.74	0.12	16.19	0.14	0.21
	KX	0.91	0.95	0.70	0.04	23.93	0.22	0.34
	QL	0.99	1.17	0.69	0.18	24.37	0.24	0.32
	RX	0.63	0.77	0.53	0.14	37.51	0.24	0.27
AN (mg/L)	BQ	0.12	0.11	0.74	-0.01	22.00	0.03	0.03
	KX	0.13	0.12	0.67	-0.01	28.71	0.04	0.05
	QL	0.12	0.12	0.65	0.00	26.96	0.03	0.04
	RX	0.13	0.13	0.84	0.00	10.97	0.01	0.02
TP (mg/L)	BQ	0.06	0.07	0.69	0.01	29.27	0.02	0.02
	KX	0.06	0.07	0.71	0.01	28.16	0.02	0.02
	QL	0.09	0.08	0.73	-0.01	23.67	0.02	0.03
	RX	0.05	0.06	0.68	0.01	29.57	0.02	0.02
Chl-a (mg/L)	BQ	0.01	0.01	0.64	0.00	32.65	0.00	0.01
	KX	0.01	0.01	0.71	0.00	30.07	0.00	0.01
	QL	0.01	0.01	0.73	0.00	28.69	0.00	0.01
	RX	0.01	0.01	0.76	0.00	25.70	0.00	0.00

Table 5. Statistical performance metrics for EFDC model validation across hydrodynamic and water quality parameters in Shanzai Reservoir.

validation accuracy achieved at BQ ($R^2 = 0.99$, $RMSE = 0.45$ m) and the consistent performance of temperature and dissolved oxygen simulations across all sampling points suggest that residual boundary condition errors did not propagate systematically into the water quality simulation results. Uncertainty in water surface abstraction is therefore considered a secondary source of model error relative to biogeochemical parameter uncertainty and sparse nutrient boundary condition data.

Validation utilized independent datasets reserved from the monitoring record, with model performance evaluated against observed water levels, temperature profiles, and water quality concentrations using multiple statistical metrics following established hydrodynamic model validation protocols⁴⁶.

Hydrodynamic validation revealed exceptional accuracy for water level simulation at BQ ($R^2 = 0.99$), confirming the model's capability to reproduce reservoir-scale hydraulic dynamics (Table 5). Surface water temperature simulations achieved relative errors ranging from 4.51% to 8.73% across sampling points, with maximum RMSE of 2.45 °C, indicating reliable representation of thermal stratification processes critical for biogeochemical cycling. Water quality parameter simulation exhibited variable performance reflecting the complexity of biogeochemical processes and data availability constraints. Dissolved oxygen simulations achieved relative errors between 14.09% and 17.87% with mean absolute differences below 2.0 mg/L, while Chemical Oxygen Demand remained within acceptable thresholds (<20% relative error). Nutrient parameters showed greater variability, with total nitrogen achieving relative errors of 16.19%–24.37% at most sampling points, though RX exhibited higher deviation (37.51%) likely reflecting complex inlet dynamics and sparse boundary condition data. Total phosphorus simulations demonstrated relative errors of 23.67%–29.57%, while chlorophyll-a ranged from 25.70% to 32.65%, both falling within acceptable ranges for eutrophication modeling applications⁴⁷.

The spatial distribution of model errors revealed systematic patterns related to hydrodynamic complexity and monitoring network coverage. Central basin sampling points (KX, QL) generally exhibited superior performance compared to inlet (RX) and outlet (BQ) locations, reflecting the challenges of simulating transitional zones with limited boundary condition data. Overall model performance achieved an average relative error of 19.66% across all parameters with mean R^2 of 0.70, demonstrating acceptable accuracy for reservoir water quality simulation consistent with international modeling standards⁴⁶.

Beyond surface comparisons, vertical temperature and dissolved oxygen profiles were evaluated at the deepest sampling location (KX, maximum depth ~ 18 m) to assess stratification representation. Observed versus modeled vertical profiles during stratified (June–September) and mixed (December–March) periods are shown in Appendix Figs. B.1, B.2. The model achieved a mean absolute thermocline depth error of 0.8 m (± 0.5 m SD) during stratified periods, indicating reasonable representation of thermal structure despite coarse vertical resolution (5 sigma layers). Modeled epilimnion-hypolimnion temperature gradients (8–12 °C) matched observed gradients (7–11 °C) within 1–2 °C, confirming adequate simulation of stratification strength.

Dissolved oxygen vertical distributions showed good agreement with observations, reproducing surface supersaturation (9–11 mg/L) and hypolimnetic depletion (4–6 mg/L) patterns (Appendix Fig. B.2). However, the model did not capture occasional near-bottom hypoxia ($\text{DO} < 2$ mg/L) observed during intense stratification events (3 out of 48 sampling dates), likely due to underestimation of sediment oxygen demand, which was set to literature values rather than site-specific measurements, and inadequate resolution of benthic boundary layer processes with only 5 vertical layers. This limited representation of near-bottom hypoxia suggests that simulated sediment phosphorus release may be underestimated during peak stratification, reinforcing our conclusion that internal nutrient loading cannot be quantitatively resolved with the current model configuration and observational dataset.

Machine learning algorithm implementation

Algorithm selection rationale

Algorithm selection was based on four critical criteria: (1) capability to handle nonlinear relationships between spectral characteristics and water quality parameters, (2) robustness to noise and outliers inherent in satellite observations, (3) computational efficiency for operational implementation, and (4) interpretability for understanding physical relationships. Based on these criteria, five complementary algorithms were implemented: Support Vector Regression (SVR), Random Forest (RF), K-Nearest Neighbors (KNN), AdaBoost (AB), and Gradient Boosting (GB).

Mathematical formulations

Support Vector Regression extends support vector machines to regression tasks by constructing an optimal hyperplane within a tolerance margin, enabling nonlinear mapping through kernel transformations:

$$f(x) = \sum_{i=1}^N \alpha_i k(x_i, x) + b \quad (4)$$

where α_i represent support vector weights, $k(x_i, x)$ denotes the kernel function enabling nonlinear transformations, and b is the bias term⁴⁸. Random Forest constructs multiple decision trees using bootstrap sampling and averages predictions to improve stability and reduce overfitting:

$$y = \frac{1}{T} \sum_{t=1}^T h_t(x) \quad (5)$$

where $\hat{y}$ represents the predicted output, T is the number of trees, and $h_t(x)$ is the prediction from the t -th decision tree⁴⁹.

K-Nearest Neighbors estimates outputs based on weighted averages of nearest training samples in feature space:

$$\hat{y}(x) = \frac{1}{K} \sum_{i=1}^K y_i \quad (6)$$

where K represents the number of nearest neighbors and y_i is the observed output of the i -th nearest neighbor⁵⁰.

AdaBoost iteratively combines weak learners by reweighting misclassified samples:

$$F(x) = \sum_{m=1}^M \alpha_m h_m(x) \quad (7)$$

where α_m is the weight assigned to the m -th learner and $h_m(x)$ represents weak learner predictions⁵¹.

Gradient Boosting fits residuals from previous learners to minimize loss functions:

$$F_m(x) = F_{m-1}(x) + \gamma_m h_m(x) \quad (8)$$

where $F_m(x)$ represents ensemble predictions after the m -th iteration and γ_m controls the learning rate⁵².

Feature engineering and model training

Spectral features included all available Landsat bands plus derived indices (NDVI, NDWI, turbidity indices) and band ratios optimized for aquatic applications²⁹. Training datasets were constructed by matching satellite acquisition dates with in situ measurements within ± 3 days, ensuring temporal consistency between observations

and imagery following established protocols for satellite-ground truth matching⁵³. The 8:2 training-validation split provided sufficient data for robust model development while maintaining independent validation capabilities.

Error analysis and performance metrics

Model performance assessment utilized comprehensive statistical metrics to evaluate accuracy across different scales and conditions. Primary metrics included Mean Error (ME), Mean Absolute Error (MAE), Relative Error (RE), Root Mean Square Error (RMSE), and coefficient of determination (R^2):

$$ME = \left(\sum_{i=1}^N (O_i - X_i) \right) / N \quad (9)$$

$$MAE = \left(\sum_{i=1}^N |O_i - X_i| \right) / N \quad (10)$$

$$RE = \left(\sum_{i=1}^N |O_i - X_i| \right) / \sum_{i=1}^N O_i \times 100\% \quad (11)$$

$$RMSE = \sqrt{\left(\sum_{i=1}^N (O_i - X_i)^2 \right) / N} \quad (12)$$

$$R^2 = 1 - \left(\sum_{i=1}^n (X_i - O_i)^2 \right) / \left(\sum_{i=1}^n (X_i - \bar{O}_i)^2 \right) \quad (13)$$

where O_i represents observed values, X_i denotes corresponding simulated values, $\bar{O}_i$ is the mean of observed values, and N represents the number of valid data points. These metrics provide complementary perspectives on model performance, with RMSE emphasizing large errors, MAE providing robust central tendency measures, and R^2 indicating explained variance, following standard practices for hydrological model evaluation⁵⁴.

Data availability

The data generated from the study will be provided by the corresponding author upon request.

Appendix A

Parameters	Value	Unit
Horizontal Eddy viscosity coefficient (Constant)	0.1	m ² /s
Horizontal momentum diffusion coefficient	0.2	-
Background vertical eddy viscosity	1×10^{-6}	m ² /s
Background molecular diffusivity	1×10^{-8}	m ² /s
Bottom roughness coefficient (Manning's n)	0.01	-
Wind sheltering coefficient	1	-
Active bed thermal layer thickness	5	M
Initial bed temperature	15	°C
Shortwave radiation rapid attenuation factor	0.45	-
Fraction of shortwave radiation absorbed at surface	0.3	-
Maximum growth rate of cyanobacteria	8	1/d
Maximum basal metabolic rate of cyanobacteria	0.01	1/d
Maximum predation (Grazing) rate on cyanobacteria	0.01	1/d
Lower optimal temperature for cyanobacteria growth	20	°C
Upper optimal temperature for cyanobacteria growth	25	°C
Maximum nitrification rate	1	1/d
Minimum hydrolysis rate of refractory particulate organic nitrogen	0.001	1/d
Minimum hydrolysis rate of labile particulate organic nitrogen	0.01	1/d
Minimum mineralization rate of dissolved organic nitrogen	0.01	1/d
Minimum hydrolysis rate of refractory particulate organic phosphorus	0.01	1/d
Minimum hydrolysis rate of labile particulate organic phosphorus	0.071	1/d

Parameters	Value	Unit
Minimum mineralization rate of dissolved organic phosphorus	0.15	l/m
Temperature coefficient for hydrolysis	0.069	-
Temperature coefficient for mineralization	0.069	-

Table A.1. EFDC parameter results after calibration.

Date	RE between inversion results and measured values (%)				RE EFDC simulation and measured values (%)			
	BQ	KX	QL	RX	BQ	KX	QL	RX
2013.07.03	5.91	1.19	-6.31	-	-12.32	20.48	169.55	-
2013.08.04	49.47	153.94	-10.99	-	-17.50	304.72	9.35	-
2013.10.15	-6.75	-33.79	-25.52	-	-1.57	13.85	-11.02	-
2013.12.02	33.94	0.05	-26.44	-	-20.12	-3.23	-18.78	-
2014.01.03	11.46	-22.51	-32.30	-	37.49	48.25	-20.84	-
2014.06.04	7.71	-6.94	-9.31	-	16.52	16.46	-21.33	-
2014.08.07	44.59	17.17	55.20	-	-2.17	-14.75	-7.16	-
2014.09.08	81.30	4.52	-9.54	-	41.88	-16.11	-6.57	-
2014.10.10	112.33	100.59	21.27	-	49.73	-7.11	10.18	-
2014.11.03	58.54	92.49	156.06	-	-71.83	-24.92	5.82	-
2014.12.05	178.54	-9.53	-27.15	-	1.36	42.90	462.95	-
2015.01.14	75.61	100.08	23.84	-	-27.40	59.33	39.30	-
2015.02.07	56.04	0.09	-28.11	-	-15.20	-2.01	32.51	-
2015.04.04	-26.46	-12.58	20.65	-	15.30	-26.70	311.53	-
2022.04.07	110.44	-17.61	-	-6.67	44.35	-24.43	-	9.92
2022.05.17	-41.21	-51.54	-	-50.44	-19.69	39.12	-	-18.86
2022.07.20	39.75	45.78	-	76.22	-3.03	29.91	-	35.44
2022.08.05	-	-1.78	-	-	-	-22.17	-	-
2022.10.24	60.18	162.70	-	70.08	373.94	144.55	-	111.89
Mean	47.30	27.49	7.24	22.30	21.65	30.43	68.25	34.60

Table A.2. The relative errors between Chl-a inversion and EFDC simulation and the measured values.

Date	RE between inversion results and measured values (%)				RE EFDC simulation and measured values (%)				
	BQ	KX	QL	RX	BQ	KX	QL	QL	BQ
2013.07.03	1.51	-21.41	18.84	-		161.81	70.07	75.95	-
2013.08.04	6.38	44.34	-9.85	-		39.53	46.81	-12.37	-
2013.10.15	31.06	32.56	38.88	-		32.40	56.42	56.50	-
2013.12.02	-5.35	-14.78	23.05	-		-1.99	12.70	15.07	-
2014.01.03	1.19	-23.17	5.80	-		-0.07	-4.83	-3.61	-
2014.06.04	49.29	28.08	-1.44	-		170.76	58.47	-36.68	-
2014.08.07	33.13	28.30	37.20	-		101.03	-0.56	29.61	-
2014.09.08	9.68	25.05	10.18	-		5.95	0.50	5.32	-
2014.10.10	40.75	14.50	8.92	-		8.53	-13.59	-0.39	-
2014.11.03	45.99	19.48	17.64	-		18.03	19.85	20.13	-
2014.12.05	-6.31	3.39	45.44	-		-5.72	57.95	45.03	-
2015.01.14	-11.67	7.78	7.39	-		-27.05	4.96	11.61	-
2015.02.07	-8.63	0.83	-14.56	-		-8.31	-1.15	4.18	-
2015.04.04	-9.19	-5.98	-18.98	-		13.04	9.97	2.12	-
2022.02.26	54.16	-2.38	-	42.21		72.80	18.99	-	92.58
2022.04.07	19.34	-25.15	-	1.22		75.08	-1.90	-	23.85
2022.05.17	36.93	26.45	-	36.96		42.11	36.57	-	30.09
2022.07.20	-14.41	18.99	-	-		22.65	25.82	-	-
2022.08.05	31.97	15.59	-	49.89		23.24	59.72	-	103.86

Date	RE between inversion results and measured values (%)				RE EFDC simulation and measured values (%)				
	BQ	KX	QL	RX	BQ	KX	QL	QL	BQ
2022.09.22	-	-54.66	-	149.22	-	-	-52.66	-	155.36
2022.10.24	19.02	-34.96	-	21.87	22.52	-28.67	-	-	59.15
Mean	16.24	3.95	12.04	50.23	38.32	17.88	15.18	-	77.48

Table A.3. The relative errors between TN inversion and EFDC simulation and the measured values.

Date	RE between inversion results and measured values (%)				RE EFDC simulation and measured values (%)				
	BQ	KX	QL	RX	BQ	KX	QL	七里	BQ
2013.07.03	21.14	4.57	6.62	-	24.89	6.67	9.15	-	-
2013.08.04	61.19	35.32	-24.66	-	87.55	46.69	-41.58	-	-
2013.10.15	60.63	75.38	84.80	-	39.07	52.58	45.81	-	-
2013.12.02	90.86	65.22	8.14	-	40.49	32.74	-11.08	-	-
2014.01.03	-6.31	3.44	34.43	-	-31.28	-22.73	0.99	-	-
2014.06.04	27.00	-1.31	-9.43	-	49.18	-0.06	-29.43	-	-
2014.08.07	11.82	9.68	15.84	-	-14.65	-7.38	-26.66	-	-
2014.09.08	29.35	35.77	-5.07	-	56.18	20.32	-7.09	-	-
2014.10.10	80.62	36.32	18.70	-	86.77	11.12	-4.81	-	-
2014.11.03	148.21	116.19	25.49	-	137.83	80.44	-15.93	-	-
2014.12.05	42.95	33.00	69.08	-	-2.97	17.76	-8.88	-	-
2015.01.14	29.70	33.60	-8.23	-	37.33	-6.01	-17.15	-	-
2015.02.07	44.10	94.42	63.87	-	-21.11	3.91	-23.08	-	-
2015.04.04	-25.62	-3.49	-22.59	-	-8.03	-18.04	-45.56	-	-
2022.02.26	-4.65	-0.19	-	58.49	225.96	30.77	-	104.51	-
2022.04.07	9.86	-12.57	-	-2.22	86.00	20.04	-	35.82	-
2022.05.17	-17.32	-0.60	-	-0.51	76.43	93.63	-	96.07	-
2022.07.20	-10.29	62.93	-	-	189.31	155.54	-	-	-
2022.08.05	-10.98	17.19	-	8.82	73.09	110.09	-	107.13	-
2022.09.22	-	4.60	-	126.73	-	299.25	-	41.68	-
2022.10.24	9.92	-13.90	-	45.48	17.14	4.37	-	75.66	-
Mean	29.61	28.36	18.36	39.46	57.46	44.37	-12.52	76.81	-

Table A.4. The relative errors between TN inversion and EFDC simulation and the measured values.

Appendix B

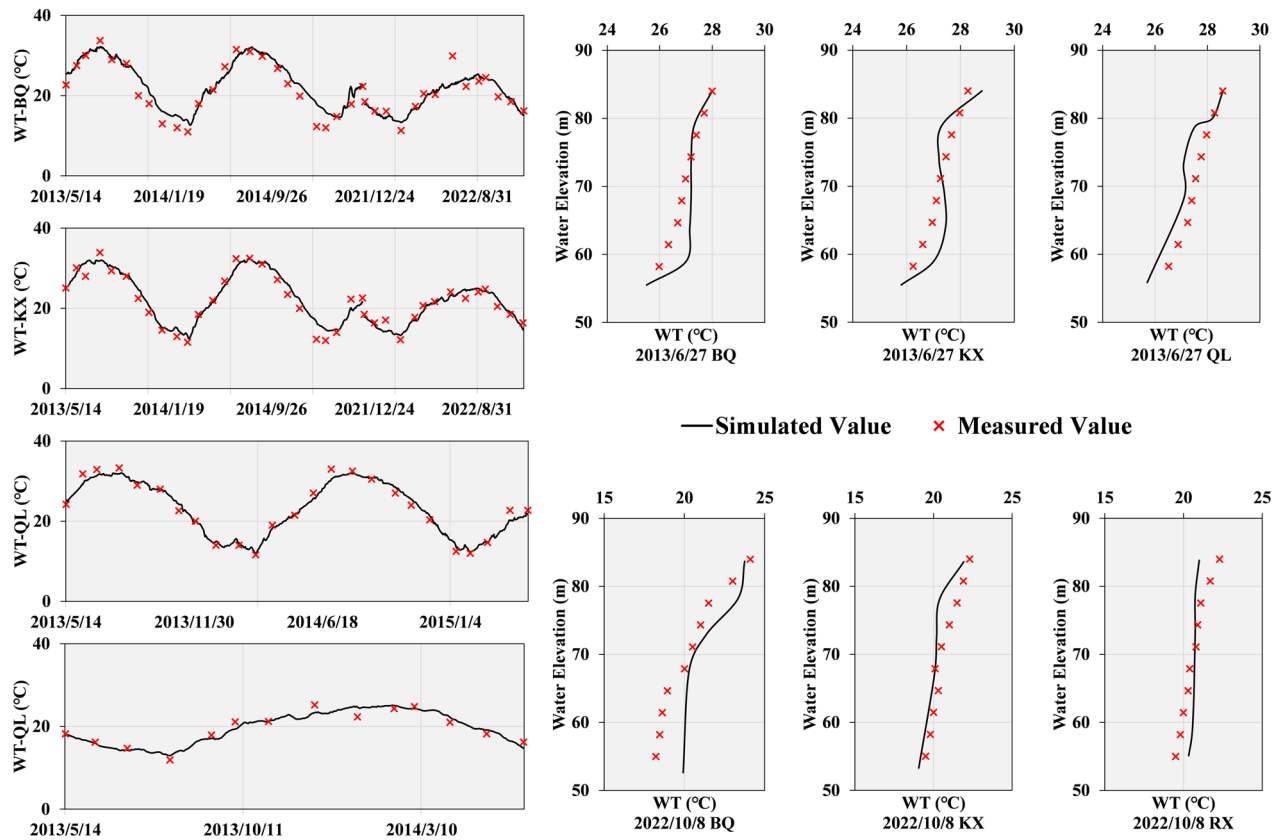


Fig. B.1. Comparison of coupled model simulations and measured values of WT.

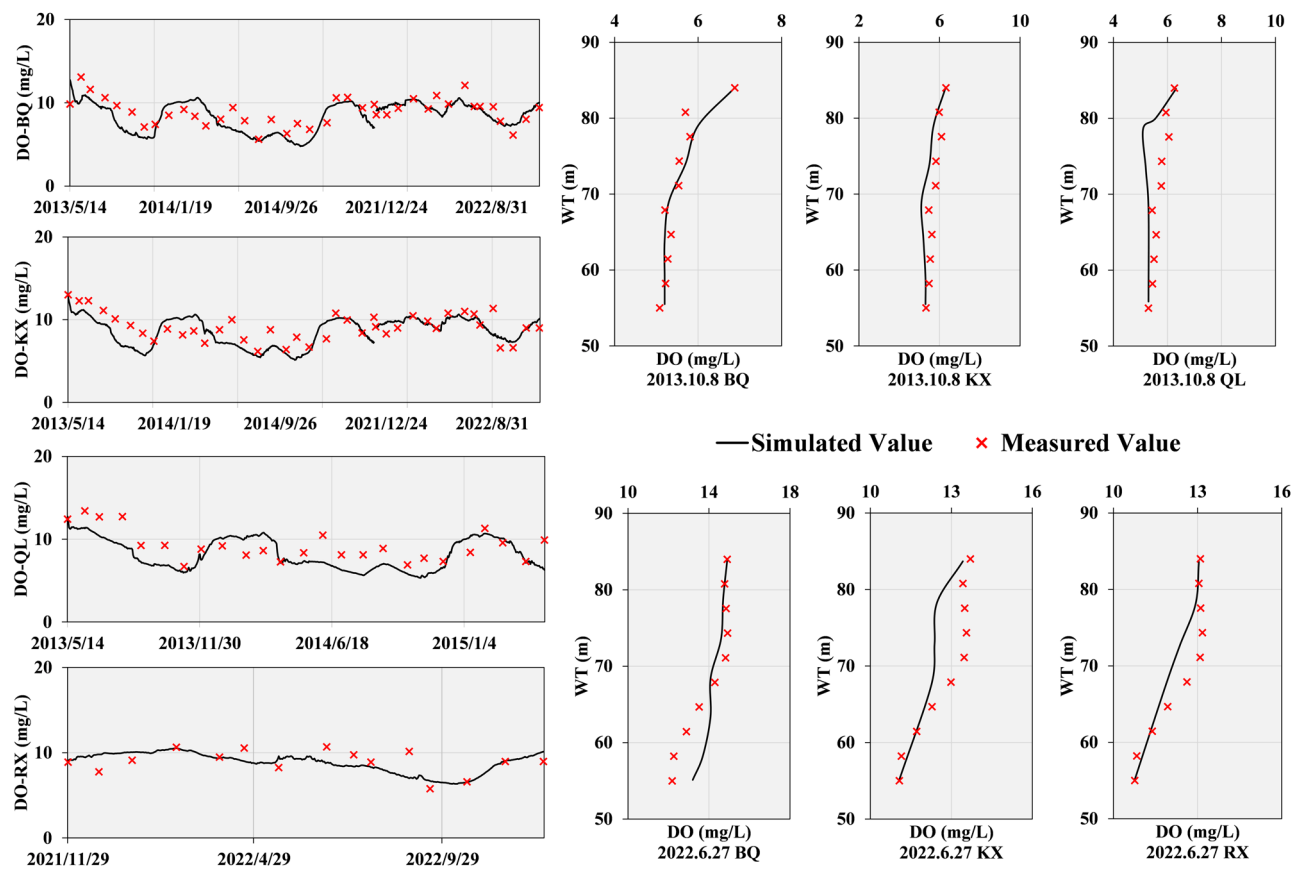


Fig. B.2. Comparison of coupled model simulations and measured values of DO.

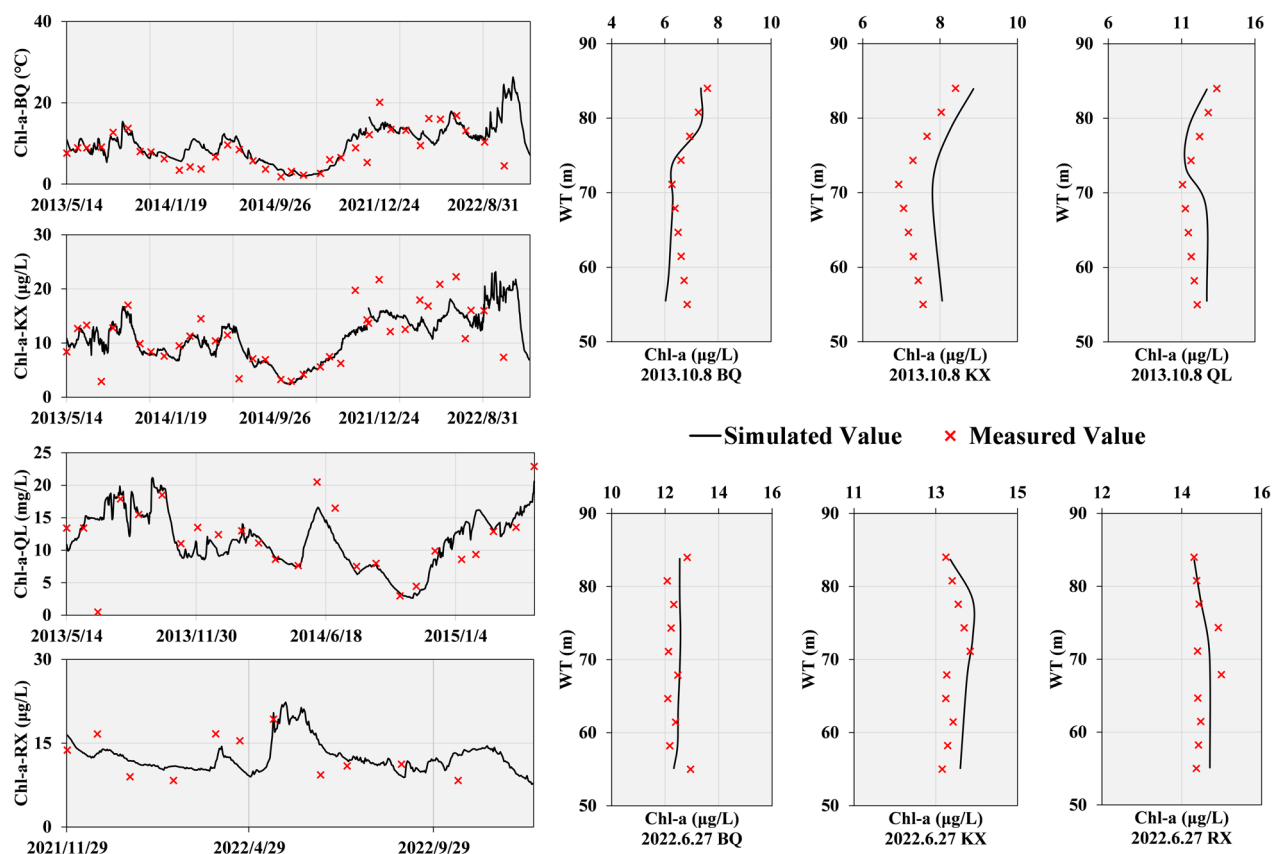


Fig. B.3. Comparison of coupled model simulations and measured values of Chl-a.

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Conceptualization, H.M. and J.Z.; methodology, H.M. and Y.C.; software, H.M., X.M. and B.L.; validation, J.Z. and X.M.; formal analysis, H.M. and Y.C.; investigation, H.M. and Z.Z.; resources, J.Z. and X.M.; data curation, H.M., Y.C. and Z.Z.; writing—original draft preparation, H.M.; writing—review and editing, H.M., J.Z. and X.M.; visualization, H.M., J.Z. and B.L.; supervision, J.Z., Z.Z. and B.L.; project administration, J.Z.; funding acquisition, J.Z., and X.M. All authors have read and agreed to the published version of the manuscript.

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Declarations

Competing interests

The authors declare no competing interests.

Additional information

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